

Euclidean Geometric Response Dictionary and Selection Rules for Four Thurston Geometries

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Abstract

We analyse Einstein–Cartan gravity with a Nieh–Yan term on the common $M^3 \times S^1$ minisuperspace and construct, across the four Thurston geometries T^3 , Nil^3 , S^3 , and Sol^3 , a *Euclidean geometric response dictionary* together with the corresponding *selection rules*. The dictionary is organised into two layers.

The first layer is the bulk layer comprising the homogeneous baseline and the inhomogeneous/nonlocal response. It is summarised by a 36-entry inventory and furnishes geometry-dependent rules for Maxwell universality, Chern–Simons activation, the Higgsing pattern, and the reduced η -mode defect-localisation benchmark. The second layer is the boundary/cobordism layer, which consists of three observable families—spectral, torsional-charge, and inflow—and a pairwise cobordism dictionary covering all six geometry pairs.

Within this structure, Nil^3 provides a strict local APS spectral entry, S^3 provides a frame-bundle-normalised torsional-charge entry, T^3 serves as the trivial spectral benchmark, and Sol^3 supports a spin-structure-dependent global spectral branch on a compact mapping-torus benchmark. Local and global sources are thereby distinguished within the spectral family itself. The resulting rules organise allowed, protected, and activated responses uniformly across the four geometries.

We further give two readings of the same Euclidean circle—a Kaluza–Klein (KK) reduction reading and a Matsubara-style thermal reading ($\beta = L$)—and show that they coexist as secondary interpretations without being identified theoretically.

1 Introduction

1.1 Background

In papers I–III/III (EC) [1, 2, 3, 4] we developed the minisuperspace of Einstein–Cartan + Nieh–Yan theory through topology-dependent phase classification, the homogeneous mode dictionary of $S^3 \times S^1$, and the three-topology comparison among T^3 , Nil^3 , and S^3 . In paper III (EC) [4] the homogeneous mode dictionary and the low-order EFT language were organised for both the Levi–Civita and the Einstein–Cartan connections, providing a baseline for the analysis of the homogeneous degrees of freedom.

These results, however, do not yet supply a single dictionary that surveys “which response is allowed, protected, and activated in which geometry” beyond the mode content of each individual geometry. In particular, in order to align inhomogeneous bulk responses, observables on boundaries/interfaces, and cobordism-type comparisons between geometry pairs in a common convention, one requires a comparison framework one level higher than the homogeneous baseline.

The aim of the present paper is to construct this comparison framework explicitly for the four Thurston [10] geometries T^3 , Nil^3 , S^3 , and Sol^3 . The focus is not on re-deriving individual modes but on presenting a response dictionary across the four geometries together with selection rules that compress its content.

1.2 Claims of the Present Paper

The central claims of this paper can be summarised in the following four points.

1. For the four Thurston geometries T^3 , Nil^3 , S^3 , and Sol^3 , the Euclidean response can be systematised in two layers: a first-layer bulk and a second-layer boundary/cobordism.
2. In the first layer a 36-entry bulk inventory closes together with geometry-dependent selection rules.
3. In the second layer the three observable families spectral/torsional-charge/inflow close, and a pairwise cobordism dictionary is given for all six geometry pairs.
4. The Euclidean circle admits two readings, the KK reduction reading and the Matsubara-style thermal reading, supplying the reduced description and the thermal description respectively.

We contrast in particular the strict local APS spectral entry of Nil^3 , the frame-bundle-normalised torsional-charge entry of S^3 , the trivial spectral benchmark of T^3 , and the spin-sensitive global spectral branch that arises on a compact Sol^3 benchmark. Although Sol^3 is inert in the CS direction count, it carries an EC slice minimum and a rigid scaffold;

in the boundary layer it forms a distinct entry that admits a spectral branch depending on the compact quotient and the spin structure.

1.3 Comparison Viewpoint

The objects under comparison are the geometric responses defined on the Euclidean product structure $M^3 \times S^1$. Concretely, we treat the homogeneous baseline of the four geometries, the inhomogeneous/nonlocal bulk response, the boundary/cobordism observables, the pairwise cobordism dictionary, and the two readings of the Euclidean circle.

The comparison admits three viewpoints. The first is the geometry-wise inventory obtained by examining each geometry in isolation. The second is the taxonomy that splits the observables into the three families spectral/torsional-charge/inflow. The third is the pairwise cobordism dictionary based on the six pairs constructed from the four geometries. Supplementing this with the geometric scaffold information allows one to distinguish the class of an observable from the distinctness of the background geometry.

1.4 Outline

The paper is organised as follows. Section 2 fixes the homogeneous baseline that supplies the common comparison axis for the four geometries. Section 3 gives the first-layer inventory of the inhomogeneous/nonlocal bulk response. Section 4 organises the boundary/interface response into a geometry-wise inventory, an observable taxonomy, a pairwise cobordism dictionary, and a geometric scaffold layer. Section 5 states the bulk–boundary selection rules as the main result. Section 6 describes the secondary readings of the Euclidean circle, the KK reduction reading and the Matsubara-style thermal reading. Section 7 discusses the place of the present paper and its future directions, and Section 8 concludes. Appendices collect notation conventions, complete inventories, KK–Matsubara coexistence tables, the normalisation chain, and derivations.

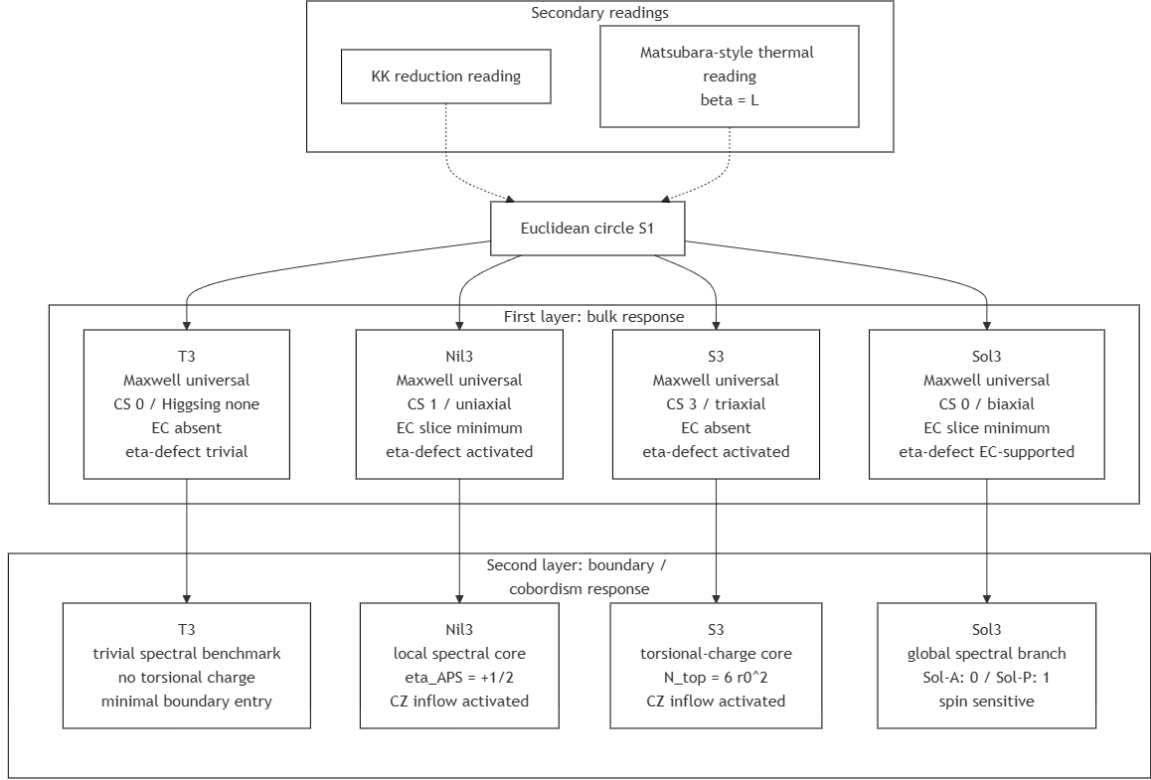


Figure 1: Two-layer architecture overview. The first-layer bulk response and the second-layer boundary/cobordism response are displayed along the four geometries, while the KK reduction reading and the Matsubara-style thermal reading—both arising from the same Euclidean circle—are separated as secondary readings.

2 Homogeneous Baseline

This section organises the homogeneous baseline of the four Thurston geometries T^3 , Nil^3 , S^3 , Sol^3 and fixes the comparison axis. The aim is to align the mode language, the baseline invariants, and the minimal/activated distinction of each geometry in a common convention before the bulk/boundary response dictionaries of the subsequent sections are introduced. The selection rules that appear later are read off from the baseline comparison fixed here.

2.1 Four-Geometry Setup

All four geometries compared in this paper share the common Euclidean product structure $M^3 \times S^1$, but the structure constants of the three-dimensional spatial part, the Weyl curvature, the KK splitting pattern, and the frame rigidity differ from geometry to geometry. Even with the same minisuperspace variables, the stability of the effective potential, the mass splitting of the vector sector, and the activation conditions of the Chern–Simons channel therefore vary in a geometry-dependent way.

We first fix the homogeneous sector as a comparison baseline, on top of which the first-layer bulk response and the second-layer boundary response are stacked. This makes the words *activated*, *protected*, and *minimal* read in subsequent sections not as descriptive labels but as clear comparative judgements relative to the homogeneous background.

2.2 Basic Comparison Table

The current baseline is summarised in the following table.

Indicator	T^3	Nil^3	S^3	Sol^3
CS direction count	0	1	3	0
KK Higgsing	none	uniaxial	triaxial	biaxial
EC slice minimum	absent	present	absent	present
Maxwell baseline	universal	universal	universal	universal
spin-2 rigidity	non-rigid	non-rigid	non-rigid	rigid
C_{LC}^2	0	$4/(3R^4)$	0	$16/(3R^4)$

Table 1: Four-geometry homogeneous baseline.

This table conveys two facts simultaneously. First, Nil^3 and S^3 are non-trivial activated geometries with respect to both the CS direction count and the Higgsing pattern. Second, Sol^3 is on the minimal side in the CS direction count but possesses a non-vanishing Weyl curvature and hence an EC slice-minimum branch isomorphic to that of Nil^3 . Accordingly Sol^3 does not belong to the inert class of T^3 ; it is a CS-inert but spin-0 active rigid

geometry. In the boundary layer this bulk distinction is reinforced by the presence or absence of spectral branches depending on the compact quotient and the spin structure.

2.3 Maxwell Universality and the CS Direction-Count Law

The first universal statement in the bulk layer is the geometry-independence of the Maxwell coefficient. Throughout this paper we adopt

$$K_{\text{Mxw}} = -\frac{L^2}{2r_0^4} \quad (1)$$

as the baseline coefficient common to the four geometries. This universality keeps the skeleton of the main dictionary intact under inhomogeneous opening and under either reading of the Euclidean circle. Geometry dependence therefore appears not in the Maxwell sector itself but in the CS-active direction count, the Higgsing pattern, the EC slice-minimum structure, and the defect response.

By contrast, Chern–Simons activation is geometry-dependent and the direction count is set by the off-diagonal structure. Activation occurs along one direction in Nil^3 and along three directions in S^3 , while in Sol^3 a self-referential cancellation forces it to zero. The distinction is decided not by “whether structure constants exist” but by “whether the correction to $\tilde{F}_{ij}$ has its leg outside the index pair $\{i, j\}$ ”.

The cancellation in Sol^3 arises because the underlying structure constants form the symmetric pair

$$C^1_{01} = +\frac{1}{R}, \quad C^2_{02} = -\frac{1}{R}, \quad (2)$$

so that only same-leg self-coupling corrections remain. Specifically the correction $A_1 \rightarrow \tilde{F}_{01}$ has its leg 1 inside $\{0, 1\}$ and is thus self-referential, whereby the contributions $+1/R$ and $-1/R$ cancel algebraically (off-diagonal CS rule, §5.1 R2; details in Appendix E.4). In this case the CS channel does not generate any new parity-odd direction; only the biaxial Higgsing remains on top of the Maxwell baseline.

2.4 Mode Dictionary and Higgsing Pattern

Read in the mode language, the differences among the four geometries in the homogeneous sector concentrate in three sectors. In the spin-0 sector, the EC slice minimum on the $\eta = V = 0$ section appears in Nil^3 and Sol^3 . Both share the same stationary radius and the same full homogeneous Hessian criterion, but the slice potential and the radial Hessian of Sol^3 are four times those of Nil^3 . T^3 has a flat zero slice and no isolated branch, while round S^3 has a non-zero slope and no stationary point on the slice (verified in Appendix E.7). In the spin-1 sector the Higgsing pattern is classified into none/uniaxial/triaxial/biaxial; this constitutes the principal comparison axis of the bulk dictionary. In the spin-2 sector

only Sol^3 exhibits frame rigidity; the other three geometries belong to the non-rigid side.

In particular, the biaxial Higgsing of Sol^3 takes the form

$$m^2(A_1) = m^2(A_2) = -\frac{L^2}{2r_0^4}, \quad m^2(A_0) = 0. \quad (3)$$

This follows from the modified field strength $\tilde{F}_{01} = F_{01} + A_1/R$, which induces a mass term for A_1 and A_2 with the same coefficient as K_{Mxw} (same-leg self-coupling), while A_0 does not appear in $\tilde{F}$ and remains massless (KK derivation in Appendix E.2). The fact that the two massive directions share the same coefficient as the Maxwell baseline shows that Sol^3 is a distinct geometry: it does not introduce a CS-active direction but carries an EC slice branch and a rigid scaffold.

The four Higgsing types “none/uniaxial/triaxial/biaxial” are realised exactly once each by $T^3, \text{Nil}^3, S^3, \text{Sol}^3$, respectively, as can be verified by direct computation of the KK mass matrices (Appendix E.6).

The role of this section is therefore to fix the baseline that supplies the comparison axis of the response dictionary. Most technical details—the Sol^3 structure constants, the CS cancellation, and the frame rigidity—are deferred to Appendices A and E.

3 Inhomogeneous / Nonlocal Bulk Response

This section treats the first-layer bulk response. The aim is to fix, in the form of an inventory and a set of selection rules, which response entries are activated, protected, or absent for the four geometries once nonlocal / inhomogeneous opening is allowed. The bulk layer closes in a setting without boundary, yet its internal structure already produces a sharp branching of the response map across the four geometries.

3.1 Definition of the Bulk Layer

By the first layer we mean the layer of responses that can be defined on the bulk side without introducing a boundary. In addition to the homogeneous baseline—the Maxwell coefficient, the Chern–Simons direction count, and the KK Higgsing pattern—it includes the presence/absence of an EC slice minimum, the activation of parity-odd transport, the reduced η -mode defect-localisation benchmark, and the opening pattern under inhomogeneous perturbations.

The complete bulk inventory has a 36-entry structure (4 geometries $\times$ 9 entries). Rather than reproducing the full table in the body of the paper, we present a compact summary table with the principal results and a corresponding rule-set reading. The full version is deferred to Appendix B.

3.2 Bulk Inventory: Overview

The principal bulk entries are as follows.

Entry	T^3	Nil^3	S^3	Sol^3
Maxwell coefficient	$-L^2/(2r_0^4)$	$-L^2/(2r_0^4)$	$-L^2/(2r_0^4)$	$-L^2/(2r_0^4)$
CS direction count	0	1	3	0
KK Higgsing type	none	uniaxial	triaxial	biaxial
EC slice minimum	absent	present	absent	present
P -linear transport	protected	activated	activated	protected
spin-2 rigidity	non-rigid	non-rigid	non-rigid	rigid
nonlinear opening	absent	absent	activated	absent
defect-localisation response (η benchmark)	trivial	activated	activated	EC-supported / CS-inert
inhomogeneous bulk response	trivial	activated	activated	CS-inert / EC-active

Table 2: Compact bulk inventory across the four geometries.

This table makes explicit that the first principal result of the paper lies precisely in the activation map of the bulk layer. While the Maxwell coefficient is fully universal, the CS direction count, Higgsing type, EC slice minimum, and defect-benchmark response differ from geometry to geometry: the essence of the bulk layer lies less in coefficient differences than in the differences in the type of allowed responses.

Read as rules, the inventory positions Nil^3 as the geometry where CS and the EC slice minimum simultaneously rise, S^3 as the most richly CS- and Higgsing-activated geometry, T^3 as an inert benchmark, and Sol^3 as a CS-inert geometry that nevertheless carries an EC slice minimum and a rigid scaffold.

3.3 Universal Channel and Activated Channel

The most important bulk-layer formula is the Maxwell baseline common to all four geometries:

$$K_{\text{Mxw}}^{T^3} = K_{\text{Mxw}}^{\text{Nil}^3} = K_{\text{Mxw}}^{S^3} = K_{\text{Mxw}}^{\text{Sol}^3} = -\frac{L^2}{2r_0^4}. \quad (4)$$

This universal formula provides a reference plane that aligns the four geometries onto a common dictionary. Geometry-dependent differences appear in the Chern–Simons activation rule and in the Higgsing classification. In Nil^3 a single off-diagonal active direction emerges, while in S^3 all three directions are activated equivalently. By contrast in Sol^3 the self-referential corrections cancel, so that

$$\text{CS}(R) = 0 \quad \implies \quad \delta_{\text{CS}} = 0, \quad \delta P = 0 \quad (5)$$

holds algebraically.

In this sense Sol^3 is not a CS-active geometry. Unlike T^3 , however, Sol^3 carries a biaxial KK splitting, frame rigidity, and an EC slice-minimum branch, so it is not a complete inert benchmark even at the bulk layer. In the boundary layer the possibility of a spectral branch depending on the compact quotient and the spin structure remains, so its CS-inertness should not be lifted directly to triviality at the second layer.

The mechanism behind this difference is geometry-specific. In Sol^3 the symmetric pair of structure constants

$$C^1_{01} = +\frac{1}{R}, \quad C^2_{02} = -\frac{1}{R} \quad (6)$$

ensures that the non-abelian cubic correction does not increase the parity-odd channel count. Specifically, the legs of both corrections $A_1 \rightarrow \tilde{F}_{01}$ and $A_2 \rightarrow \tilde{F}_{02}$ lie *inside* the index pairs $\{0, 1\}$ and $\{0, 2\}$ respectively, so that no off-diagonal CS direction is generated for any profile-local scalar coefficients c_{01}, c_{02} (off-diagonal CS rule; see Appendices E.4 and E.5). In Nil^3 , on the other hand, a non-trivial structure constant exists in only one direction; this uniaxiality is the origin of both the one-direction CS activation and the EC slice-minimum branch. In S^3 the isotropic coupling structure preserves a three-fold degeneracy and admits a nonlinear opening through higher-order couplings.

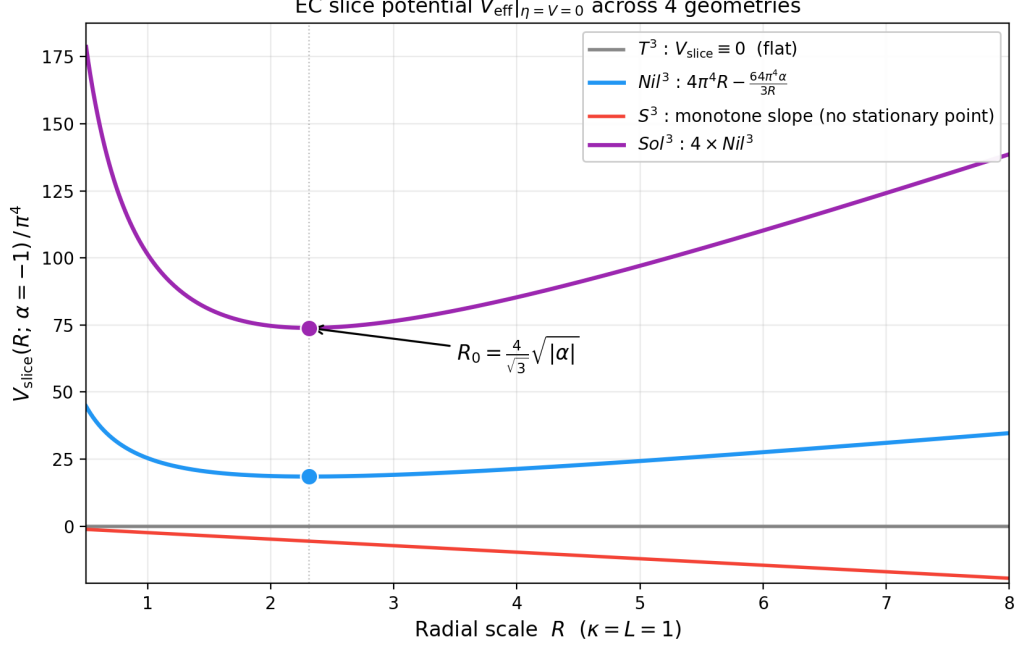


Figure 2: Comparison of the EC slice potential $V_{\text{eff}}(\eta = V = 0, R)$ across the four geometries. In Nil^3 and Sol^3 a stationary branch appears, with the slice potential of Sol^3 being four times that of Nil^3 . T^3 has a flat zero slice and round S^3 has a non-zero slope without a stationary point.

3.4 Reduced η -Mode Defect-Localisation Benchmark

To compare the four geometries on equal footing when an inward radial dip or localised defect is opened, we adopt the reduced 1D AX-torsion ansatz

$$\eta = \eta(z) \tag{7}$$

on a slice-wise homogeneous background. Setting the radial scale as

$$\rho(z) = r(z) \quad (T^3, S^3), \quad \rho(z) = R(z) \quad (\text{Nil}^3, \text{Sol}^3), \tag{8}$$

the localisation benchmark operator reduces to the Sturm–Liouville form

$$-\frac{d}{dz} \left[K_{\text{geo}}(z) \frac{df}{dz} \right] + M_{\text{geo}}(z) f = E f, \quad K_{\text{geo}}(z) = M_{\text{geo}}(z) = c_{\text{geo}} \rho(z). \tag{9}$$

The coefficient c_{geo} is topology-dependent:

$$c_{T^3} = c_{\text{Nil}^3} = c_{\text{Sol}^3} = \frac{96\pi^4 L}{\kappa^2}, \quad c_{S^3} = \frac{12\pi^2 L}{\kappa^2}. \tag{10}$$

Here M_{geo} is the exact homogeneous datum

$$M_{\text{geo}} = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \eta^2} \right|_{\eta=0} \quad (11)$$

obtained from the current `dppu` library, and K_{geo} is the reduced 1D datum coming from the AX contortion gradient; the agreement of the two is verified in Appendix E.8.

For a Gaussian ρ -dip

$$\rho(z) = \rho_0 \left(1 - A e^{-z^2/w^2} \right), \quad 0 < A < 1, \quad (12)$$

a variational evaluation with the trial function $f_\lambda = e^{-\lambda|z|}$ yields $E_{\text{var}}(\lambda) < M_0 = c_{\text{geo}}\rho_0$ for sufficiently small $\lambda > 0$, so that the existence of at least one bound state is established analytically (Appendix E.8). Localisation itself therefore holds at the level of this reduced benchmark for all four geometries. Representative numerical scans corroborate this: $E_{\text{min}} < M_0$ holds across all $4 \times 3 = 12$ examples (4 geometries $\times$ 3 Gaussian ρ -dip profiles); see scripts `paper04/eta_defect_coefficients.py` and `paper04/defect_localization.py`.

Reduced 1D η -mode defect localization

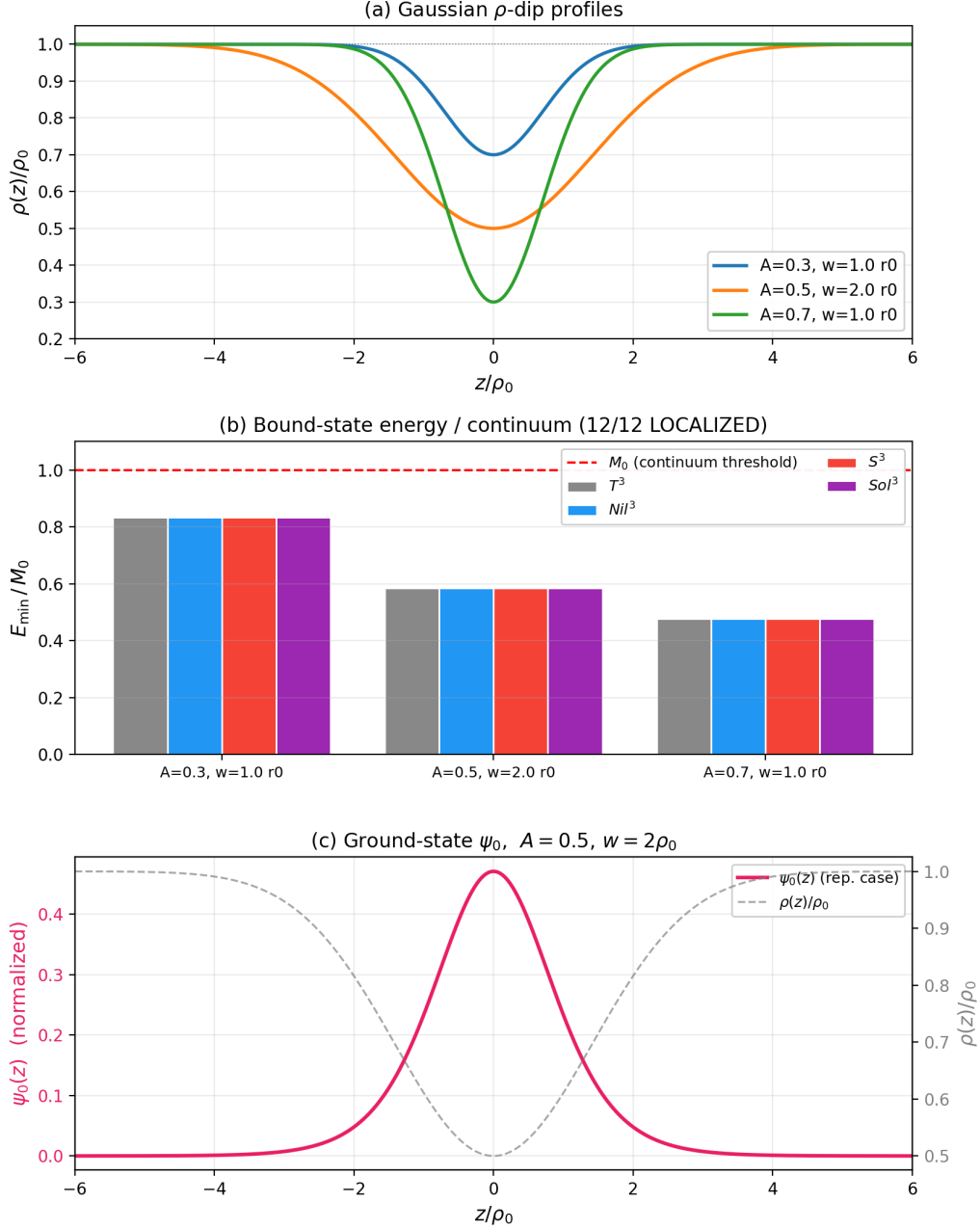


Figure 3: Reduced η -mode defect-localisation benchmark. The Gaussian ρ -dip, the effective potential $M_{\text{geo}}(z)$, the bound-state energy $E_{\min} < M_0$, and the ground-state wavefunction are displayed within the same 1D benchmark. Localisation itself is common to the four geometries; differences in activation are determined by the subsequent CS / EC / nonlinear structure.

In T^3 this common benchmark stays on the kinematic / trivial side. In Sol^3 the EC slice minimum is present so that the spin-0 sector is active, but the CS direction count remains zero so that the response does not progress to torsional-charge or local parity-odd activation. In Nil^3 , by contrast, the EC slice minimum and the CS-active structure

combine so that the localised η -mode is tied to the activation of the spectral / parity-odd channel. In S^3 the triplet degeneracy and the nonlinear opening overlap, producing the richest activation pattern. Hence the labels “trivial / activated” in the row record not the existence of a bound state *per se*, but how the common reduced η -benchmark combines with the additional bulk structure.

3.5 Geometry-Wise Bulk Reading

The bulk-layer reading splits across the four geometries as follows.

- T^3 : flat and inert. The response is minimal and many bulk entries are trivial / protected. It serves as the reference geometry for the comparison dictionary.
- Nil^3 : the first activated geometry. It carries CS-active structure, an EC slice minimum, and a uniaxial Higgsing, and connects naturally to the strict APS spectral core appearing on the boundary.
- S^3 : the richest activated geometry. It exhibits isotropic degeneracy, an activated parity-odd channel, and a nonlinear opening, and yields a frame-bundle-normalised torsional-charge core on the boundary.
- Sol^3 : a distinct geometry that is CS-inert yet carries an EC slice minimum, a biaxial KK splitting, and frame rigidity. In the boundary layer it can support a global spectral branch depending on the compact quotient and the spin structure.

3.6 Summary

The significance of the bulk layer is that it closes the response inventory of the four geometries in the form of first-layer selection rules. The organisation obtained here is reused, in the subsequent boundary/cobordism layer, as the comparison axis for deciding “which channel is non-trivial in which geometry”.

4 Boundary / Interface Response

This section addresses the second-layer boundary/cobordism response. We present the geometry-wise inventory, the observable taxonomy, the pairwise cobordism dictionary, and the geometric scaffold layer as a single structure. While the bulk layer is mainly concerned with “what each geometry allows in isolation,” the second layer makes explicit “what survives as an observable on the boundary” and “which observables make geometry pairs comparable”.

4.1 CZ Inflow and Notation

In the second layer at least three different 3-forms must be carefully distinguished:

- $\mathcal{C}_3^{\text{tor}}$: torsion-CS 3-form,
- $\mathcal{C}_3^\sigma$: spinor CS 3-form,
- $\mathcal{C}_3^{\text{adj}}$: adjoint CS 3-form.

We write the Chandia–Zanelli inflow [7] as

$$S_{\text{CZ}}[M_3] = \frac{k_{\text{cl}}}{4\pi^2} \int_{M_3} \mathcal{C}_3^{\text{tor}}. \quad (13)$$

The relation

$$\text{NY} = d\mathcal{C}_3^{\text{tor}}, \quad d\mathcal{C}_3^{\text{tor}} = T^a \wedge T_a + R_{ab} \wedge e^a \wedge e^b \quad (14)$$

between the Nieh–Yan density and $\mathcal{C}_3^{\text{tor}}$ then connects the boundary inflow directly to NY. By contrast the APS spectral entry [8] is defined via $\mathcal{C}_3^\sigma = \text{Tr}_\sigma(\omega \wedge d\omega + \frac{2}{3}\omega^3)$, constructed only from the Levi–Civita connection; the two forms must not be conflated. The adjoint CS 3-form $\mathcal{C}_3^{\text{adj}}$ appears at intermediate stages of the APS argument but is not the final spectral observable itself (Appendix E.10).

This distinction is essential for the taxonomy that follows. The strict APS entry of Nil^3 and the frame-bundle-normalised torsional-charge entry of S^3 are both parity-odd boundary quantities, but they belong to different families. Even within the spectral family, moreover, the source is not uniform: the strict value of Nil^3 is a local spectral core provided by a non-zero Levi–Civita spinor CS integral, whereas the APS branch on the compact Sol^3 benchmark is a global spectral branch that arises from kernel/spin-structure data while the local CS integral itself vanishes.

4.2 Geometry-Wise Boundary Inventory

The geometry-wise boundary inventory is summarised as follows.

Entry	T^3	Nil^3	S^3	Sol^3
CS interface jump	absent	present	present	absent
APS η -invariant	0	strict $+1/2$ (local)	0	0/1 (Sol-A / Sol-P)
CZ inflow	absent	activated	activated	absent
edge / interface mode	minimal	activated	activated	spin-sensitive
boundary tag	trivial benchmark	APS local core	torsional-charge active	global spectral branch

Table 3: Geometry-wise boundary inventory.

The principal entry of Nil^3 is the strict APS spectral core, with canonical value

$$\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = +\frac{1}{2}. \quad (15)$$

The Heisenberg mode decomposition shows that, with PPA spin structure, the condition $p_2 \in \mathbb{Z} + 1/2$ excludes the $p_2 = 0$ zero mode, so that the $n = 0$ spin-up branch alone carries the spectral asymmetry. The value itself is obtained from the Levi–Civita spinor Chern–Simons integral of Nil^3 ,

$$\int_Y \mathcal{C}_3^\sigma = -\frac{1}{2}, \quad (16)$$

combined with the DPPU orientation convention $\eta_{\text{APS}}^{(3D)} = -\int_Y \mathcal{C}_3^\sigma$, giving

$$\eta_{\text{APS}}^{(3D)} = +\frac{1}{2}, \quad \eta(0) = 2\eta_{\text{APS}}^{(3D)} - h = 1 \quad (17)$$

(Appendix E.12; verification: `paper04/eta_aps_nil3.py`). Crucially, this spectral value originates from Levi–Civita spinor spectral data rather than from the torsion-CS sector. Although Nil^3 also carries CZ inflow, the strict core that takes precedence is the spectral family, and its source is local. As benchmarks, the PPA spin structure on T^3 and the Levi–Civita Dirac on round S^3 both have spectra exactly symmetric under $\lambda \leftrightarrow -\lambda$, giving $\eta_{\text{APS}}^{(3D)} = 0$ (verification: `proofs/aps_zero_t3_s3.py`).

Nil^3 APS spectral core: $\eta_{\text{APS}}^{(3D)} = +1/2$

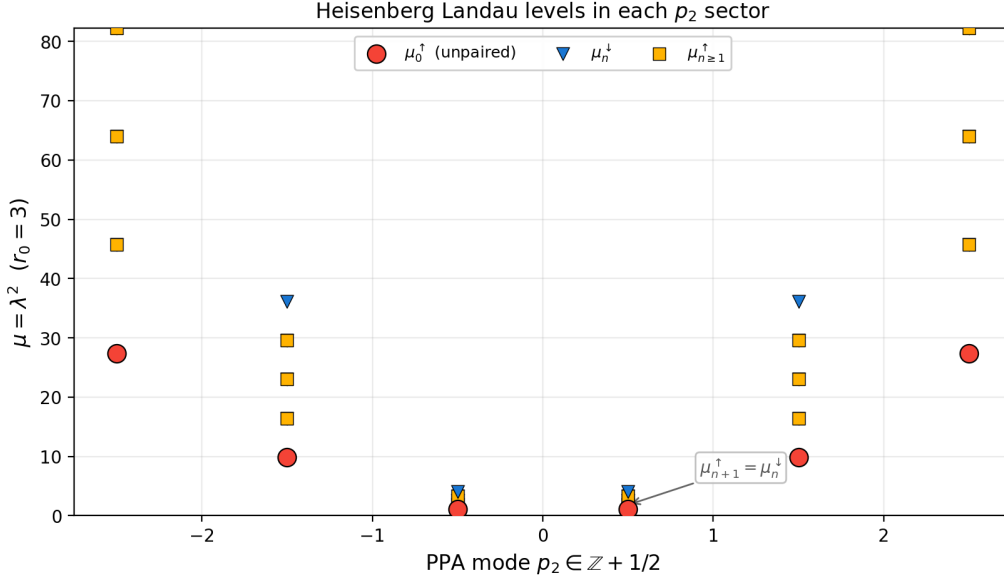


Figure 4: Nil^3 Heisenberg Landau-level spectrum and the construction of $\eta_{\text{APS}}^{(3D)}$. Under the PPA spin structure the $n = 0$ spin-up branch carries the spectral asymmetry, and the Levi-Civita spinor CS integral gives $\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = +1/2$.

For Sol^3 the situation is different. Taking the compact mapping-torus benchmark

$$M_A = T^2 \rtimes_A S^1, \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad (18)$$

together with the two spin structures Sol-P and Sol-A tracked within DPPU, the local Levi-Civita Chern-Simons integral itself vanishes:

$$\int_Y \mathcal{C}_3^\sigma = 0. \quad (19)$$

Here “Sol-P” denotes the benchmark with periodic base circle and periodic fibre, and “Sol-A” the benchmark with anti-periodic base circle and periodic fibre. The Dirac kernel branches as

$$h(\text{Sol-P}) = 2, \quad h(\text{Sol-A}) = 0, \quad (20)$$

and the antilinear time-reversal symmetry gives $\eta(0) = 0$, so that

$$\eta_{\text{APS}}^{(3D)}(\text{Sol-P}) = 1, \quad \eta_{\text{APS}}^{(3D)}(\text{Sol-A}) = 0 \quad (21)$$

(Appendix E.13; verification: `proofs/eta_aps_sol3.py`). Sol^3 is therefore not a “local APS core” but a global spectral branch that switches on/off according to the compact quotient and the spin structure.

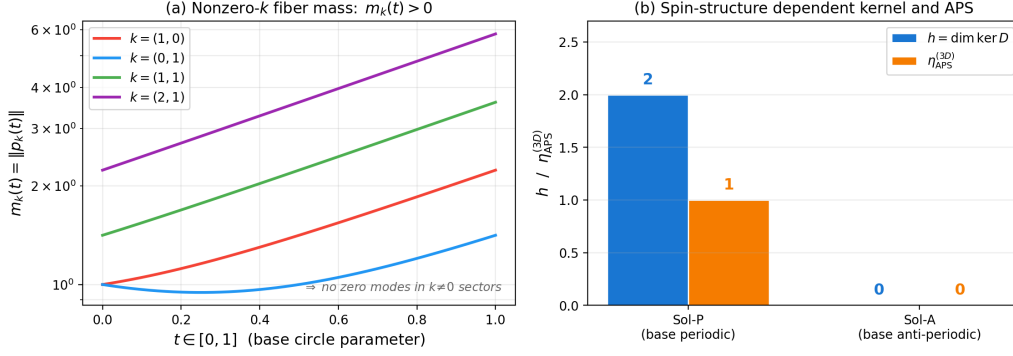


Figure 5: Global spectral branch of Sol³. On the compact mapping-torus benchmark the kernel dimension h and $\eta_{\text{APS}}^{(3D)}$ branch between Sol-P and Sol-A; the on/off switching is driven by global / spin-structure data rather than by the local CS integral.

The principal entry of S^3 is the frame-bundle-normalised torsional-charge core characterised by

$$N_{\text{top}} = 6r_0^2. \quad (22)$$

The geometric part follows from the structure constants $C^i_{jk} = (4/r_0)\epsilon^i_{jk}$ via $T^i = \frac{1}{2}C^i_{jk}e^j \wedge e^k$ and

$$\mathcal{C}_3^{\text{tor}} = e_i \wedge T^i = \frac{12}{r_0} \text{vol}_{S^3} \implies \int_{S^3} \mathcal{C}_3^{\text{tor}} = 24\pi^2 r_0^2. \quad (23)$$

The factor $1/(4\pi^2)$ is the frame-bundle normalisation associated with $\pi_3(SO(3)) = \mathbb{Z}$ ($T_R = 1$), leading to $N_{\text{top}} = 6r_0^2$ (Appendix E.11). Since no extra quantisation condition on r_0 is imposed, N_{top} here is the frame-bundle-normalised torsional charge. The interface CS jump and the CZ inflow belong to this torsional sector and underpin the role of S^3 in the pairwise dictionary as the torsional-charge-led geometry.

T^3 has all boundary entries trivial and serves as a benchmark. Yet Sol³ is CS-inert while still carrying an EC slice minimum and a rigid scaffold; in the boundary layer it can support a global spectral branch depending on the compact quotient and the spin structure. Collapsing T^3 and Sol³ into a single second-layer class is therefore inaccurate. Independently of the presence/absence of a boundary spectral branch, the fact that Sol³ has a distinct, non-flat background geometry is preserved.

4.3 Observable Taxonomy

We classify the second-layer observables into the following three families.

Family	Representative	Carrier	Value type	Canonical geometry
spectral	$\eta_{\text{APS}}^{(3D)}$	LC spectral data ($\mathcal{C}_3^\sigma$ and/or kernel/spin)	half-int. / int. branch	Nil^3 (local), Sol^3 (global)
torsional-charge	N_{top}	frame-bundle normalisation of $\mathcal{C}_3^{\text{tor}}$	frame-bundle-norm. torsional charge	S^3
inflow	S_{CZ}	$\mathcal{C}_3^{\text{tor}}$	boundary functional	Nil^3, S^3

Table 4: Three observable families of the second layer.

These three families furnish the observable taxonomy of the second layer. The crucial point is that, although $\eta_{\text{APS}}^{(3D)}$ and N_{top} are both non-trivial boundary quantities, they are not of the same type and should not be subtracted directly. Quantities universally comparable in pairwise comparison reside instead on the side of the inflow functional written in terms of $\mathcal{C}_3^{\text{tor}}$.

4.4 Pairwise Cobordism Dictionary

Six pairs are obtained from the four geometries. We summarise the dominant observable and source relation for each pair.

Pair	Class	Dominant observable	Note
$S^3 \leftrightarrow T^3$	torsional / trivial	frame-bundle-norm. torsional charge	S^3 benchmark
$S^3 \leftrightarrow \text{Nil}^3$	torsional / local spectral	mixed	torsional-charge crosses local spectral
$S^3 \leftrightarrow \text{Sol}^3$	torsional / global spectral	spin-sensitive	torsional-led for Sol-A; mixed for Sol-P
$T^3 \leftrightarrow \text{Nil}^3$	trivial / local spectral	spectral	Nil^3 local core dominates
$T^3 \leftrightarrow \text{Sol}^3$	trivial / global spectral	spin-sensitive	Sol-A: trivial; Sol-P: global spectral
$\text{Nil}^3 \leftrightarrow \text{Sol}^3$	local / global spectral	spectral	source comparison within spectral family

Table 5: Pairwise cobordism dictionary.

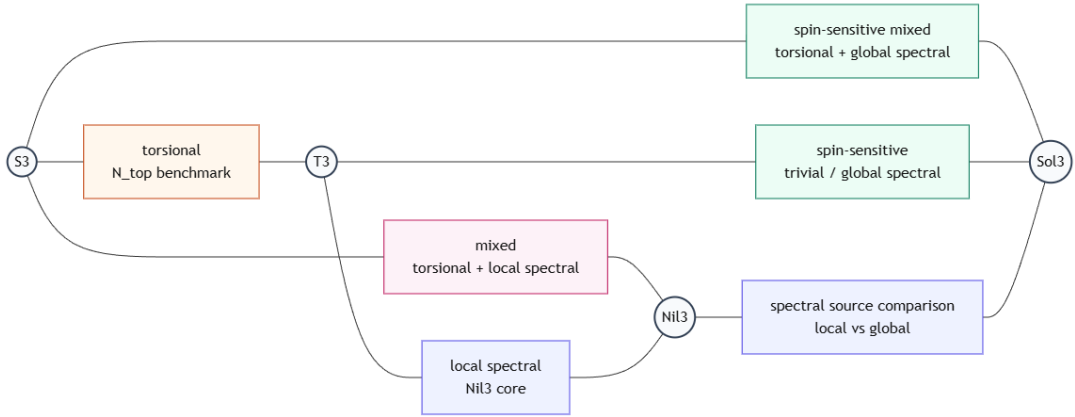


Figure 6: Pairwise cobordism dictionary. The four geometries form the vertices of a pair graph; the six pairs are labelled with the dominant observable and the spin-sensitivity. Pairs containing S^3 tend to be torsional-charge-led, whereas pairs containing Sol^3 admit a global spectral branch governed by the compact quotient and the spin structure.

This table captures the comparison logic of the paper most directly: the geometry-wise inventory translated into the language of pairwise cobordism. In particular, $S^3 \leftrightarrow \text{Nil}^3$ is the intersection of the torsional-charge family and the local spectral family, while

$\text{Nil}^3 \leftrightarrow \text{Sol}^3$ encodes a “local vs. global” source distinction within the spectral family itself. $T^3 \leftrightarrow \text{Sol}^3$ is not a uniformly inert pair either: it overlaps the trivial benchmark for Sol-A, while a global spectral branch arises for Sol-P, making it a spin-sensitive pair.

The advantage of the pairwise dictionary is that it makes simultaneously explicit “which observable family is the subject of comparison” and “whether the value is local or global”—features hard to read off from a geometry-by-geometry inventory alone. The tendency that pairs containing S^3 are torsional-charge-led, those containing Nil^3 are local-spectral-led, and those containing Sol^3 may carry a global spectral branch depending on the compact quotient and spin structure underpins the concise statement of the selection rules.

4.5 Geometric Scaffold Layer

While not an observable family, we also record the following scaffold-layer quantities, which characterise the background geometry.

Geometry	C_{LC}^2	spin-2 rigidity tag	A_{KK}/K^2
T^3	0	trivial	0
Nil^3	$4/(3R^4)$	non-rigid	$2/3$
S^3	0	non-rigid	0
Sol^3	$16/(3R^4)$	rigid	$2/3$

Table 6: Geometric scaffold quantities.

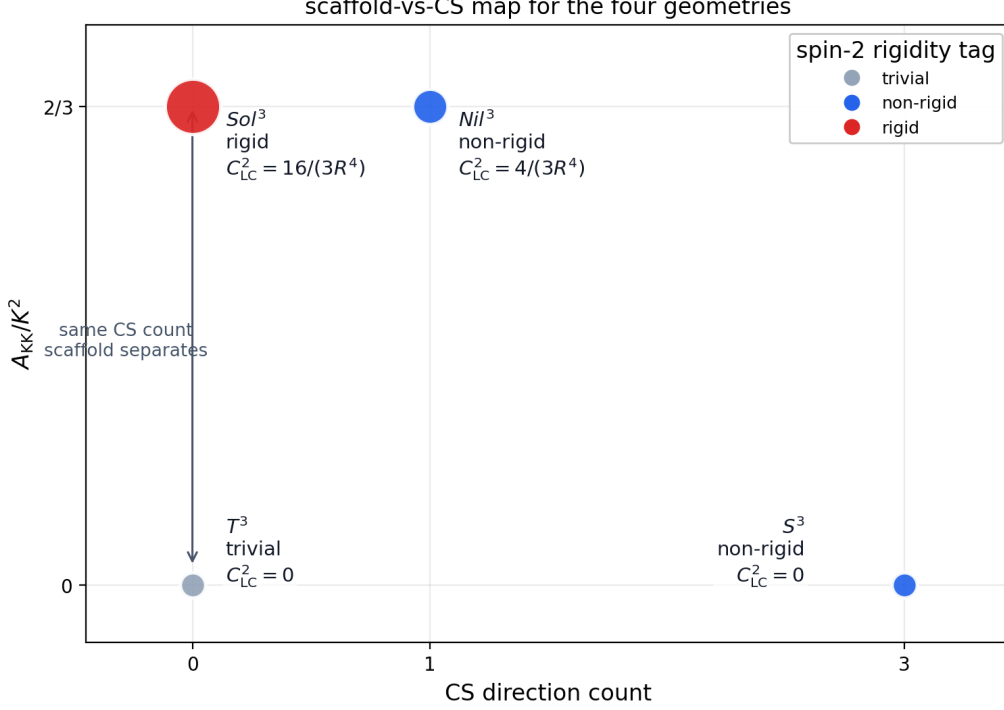


Figure 7: Scaffold-vs-CS scatter. The horizontal axis is the CS direction count, the vertical axis is A_{KK}/K^2 , the marker size encodes C_{LC}^2 , and the colour encodes the spin-2 rigidity tag. T^3 and Sol^3 share CS count 0 but separate in the scaffold layer.

The spin-2 rigidity tag is not a strict numerical invariant but a scaffold label summarising the zero pattern of the squash/shear Hessian. “Rigid” refers to the Sol^3 frame rigidity

$$\frac{\partial^2 V}{\partial \varepsilon^2} = \frac{\partial^2 V}{\partial s^2} = \frac{\partial^2 V}{\partial \varepsilon \partial s} = 0, \quad (24)$$

“trivial” to the flat kinematic zero of T^3 , and “non-rigid” to the non-zero spin-2 Hessian of Nil^3 / S^3 .

This Hessian zero pattern is a non-trivial consequence of the fact that the volume-preserving deformation $f_1 = (1+\varepsilon)^{2/3}(1+s)$, $f_2 = (1+\varepsilon)^{-2/3}/(1+s)$ of Sol^3 leaves the structure constants independent of ε, s (Appendix E.1); the script `proofs/sol3_structure.py` verifies that the three second derivatives vanish directly.

The KK anisotropy is defined as

$$A_{KK} := \sum_{i=0}^2 (m_i^2 - \bar{m}^2)^2, \quad \bar{m}^2 := \frac{1}{3} \sum_{i=0}^2 m_i^2, \quad (25)$$

normalised by $K^2 := K_{M_{\text{XW}}}^2$; we record $A_{KK}/K^2 = 0, 2/3, 0, 2/3$ as the scaffold datum.

The role of this layer is not to add a new observable family but to explain “why the activation pattern of the three families differs across the geometries”. Both T^3 and Sol^3 lack a local CS core, yet they are separated in the bulk layer by the presence/absence of an EC slice minimum and a rigid scaffold, and they are further distinguished in the

boundary layer by the global spectral branch dependent on the compact quotient and spin structure. On the Sol-A benchmark a trivial spectral profile coincident with T^3 appears, whereas the Sol-P benchmark raises a global spectral branch with $\eta_{\text{APS}}^{(3D)} = 1$. The geometric distinction—that T^3 is flat and inert while Sol^3 carries a non-vanishing Weyl curvature, an EC slice minimum, a biaxial KK splitting, and frame rigidity—persists nonetheless. This distinction does not amount to introducing further observable families, but it is indispensable for a complete characterisation of the geometries.

The second-layer dictionary thus closes through the three views: geometry-wise, pair-wise, and scaffold. In the next section we combine these with the bulk layer and compress the contents into selection rules instead of an enumeration of individual computations.

5 Selection Rules

This section compresses the results of §§2–4 into selection rules. What matters is less the detail of individual computations than a rule set surveying “which response is allowed, protected, or activated in which geometry”. The selection rules are not a restatement of the inventory; they translate the comparison of the four geometries into reusable decision rules.

5.1 Bulk Rules

The principal rules of the bulk layer can be summarised as follows. The geometric origin of each rule is collected first in the following table.

Rule	Geometric source
R1	geometry-independence of the Maxwell action
R2	presence/absence of off-diagonal structure constants
R3	mass-splitting pattern of the KK vector sector
R4	EC slice-minimum branch in the scalar potential
R5	localisation well opened by the reduced 1D η -benchmark and its activation conditions

Table 7: Geometric source of each bulk rule.

R1: Maxwell universality. The Maxwell baseline is common to the four geometries:

$$K_{\text{Mxw}} = -\frac{L^2}{2r_0^4} \quad (26)$$

is the master coefficient for all four. Geometry dependence is therefore not in the kinetic baseline but on the side of the activated channels.

Justify. The principal part of Maxwell originates from the abelian quadratic term $F_{ij}F^{ij}$, while the structure constants enter only the lower-order mass / CS terms via the modified field strength. Under the common product-circle convention, the principal coefficient is therefore fixed to $-L^2/(2r_0^4)$ for all four geometries (Appendix E.3; verification: `proofs/kk_higgsing.py`).

R2: CS activation rule. The Chern–Simons channel is activated only when the correction to $\tilde{F}_{ij}$ has its leg outside $\{i, j\}$. Nil³ (one direction) and S^3 (three directions) follow this rule; the self-referential correction of Sol³ cancels exactly.

Justify. Classifying the correction type into “no structure constant / off-diagonal / self-referential” fixes the CS direction count algebraically as 0/1/3/0 (Appendix E.5). In Sol³ the legs of $A_1 \rightarrow \tilde{F}_{01}$ and $A_2 \rightarrow \tilde{F}_{02}$ lie inside the field-strength index pair, so the CS coefficient remains zero for any profile-local coefficients $c_{01}(z), c_{02}(z)$ (verification:

`proofs/cs_cancellation.py`). This is a statement about the profile-local reduced KK ansatz, not the derivation of the full derivative-dependent inhomogeneous field equation.

R3: Higgsing classification. The KK Higgsing pattern is classified into the four types none / uniaxial / triaxial / biaxial; in this order each is realised exactly once by T^3 , Nil^3 , S^3 , Sol^3 , providing the most direct compression of the bulk-sector mode dictionary.

Justify. Direct computation of the KK mass dictionaries gives massive counts 0, 1, 3, 2, exactly exhausting the four types (Appendix E.6).

R4: EC slice-minimum rule. A horizontal evaluation of the homogeneous EC + NY + Weyl potential of the current DPPU library across the four geometries shows that an EC slice minimum on the $\eta = V = 0$ section exists in Nil^3 and Sol^3 and is absent in T^3 and round S^3 . Hence Nil^3 carries the spin-0 entry where CS and EC simultaneously rise, while Sol^3 carries the CS-inert but EC-active rigid entry.

Justify. Paper III (EC) [4] shows analytically

$$r_0 = \frac{4\kappa}{\sqrt{3}}\sqrt{|\alpha|}, \quad |\kappa^2\theta_{\text{NY}}| < 1 \Rightarrow \text{full homogeneous local minimum.} \quad (27)$$

For Sol^3 the same r_0 and the same Hessian criterion hold, with the slice potential and H_{RR} four times those of Nil^3 . T^3 has no isolated radial branch on the flat zero slice, and round S^3 has non-zero slope and no stationary point (Appendix E.7; verification: `paper04/ec_slice_minima.py`).

R5: reduced η -benchmark localisation rule. On the reduced 1D AX-torsion benchmark over a slice-wise homogeneous background, the Gaussian radial dip produces a localisation well in all four geometries. Progress to a local parity-odd activated response, however, requires CS-active structure. Hence Nil^3 is activated by CS+EC, S^3 by CS+triplet/nonlinear opening, T^3 is kinematic / trivial, and Sol^3 is EC-supported but CS-inert.

Justify. The current `dppu` library yields the exact homogeneous datum

$$M_{\text{geo}} = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \eta^2} \right|_{\eta=0} = c_{\text{geo}} \rho, \quad (28)$$

and the reduced 1D kinetic datum from the AX contortion gradient agrees: $K_{\text{geo}} = c_{\text{geo}} \rho$, with $c_{T^3} = c_{\text{Nil}^3} = c_{\text{Sol}^3} = 96\pi^4 L/\kappa^2$ and $c_{S^3} = 12\pi^2 L/\kappa^2$. The variational evaluation for a Gaussian ρ -dip then guarantees at least one bound state analytically for all four geometries, and the representative benchmark scan confirms LOCALIZED in all $4 \times 3 = 12$ examples (Appendix E.8; verification: `paper04/eta_defect_coefficients.py`, `paper04/defect_localization.py`). The differences in response are determined not by

localisation itself but by whether CS activation (R2), the EC slice minimum (R4), or the triplet/nonlinear opening is present in the subsequent stage: Nil^3 has CS+EC, S^3 has CS+triplet, Sol^3 has EC alone, and T^3 has none.

The bulk rules above show that the four geometries can be compared in the form “universal baseline + geometry-dependent activation pattern”. In particular, Sol^3 does not introduce a CS-active family but, since it carries an EC slice minimum, is not placed in the same minimal bulk class as T^3 . Sol^3 remains as the CS-inert, EC-active, rigid geometry.

5.2 Boundary Rules

The principal rules of the boundary/cobordism layer are as follows.

B1: CZ inflow activation rule. The CZ inflow is activated only in CS-active geometries. The torsion-CS sector vanishes in T^3 and Sol^3 and a non-trivial inflow remains only in Nil^3 and S^3 .

Justify. $S_{\text{CZ}} \propto \int \mathcal{C}_3^{\text{tor}}$ is non-zero only when the bulk CS is active, in one-to-one correspondence with the CS direction count 0/1/3/0 of R2.

B2: local spectral-core rule. A strict local APS spectral core appears representatively in Nil^3 with canonical value

$$\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = +\frac{1}{2}. \quad (29)$$

This entry belongs to the spinor spectral family and must not be conflated with the torsion-CS family.

Justify. The Heisenberg mode decomposition shows $h = 0$ for PPA and that only the $n = 0$ spin-up branch carries the asymmetry. The value is fixed by the Levi–Civita spinor CS integral $\int_Y \mathcal{C}_3^\sigma = -1/2$ together with the orientation convention $\eta_{\text{APS}}^{(3D)} = -\int_Y \mathcal{C}_3^\sigma$, giving $\eta_{\text{APS}}^{(3D)} = +1/2$ and $\eta(0) = 1$ (Appendix E.12; verification: `paper04/eta_aps_nil3.py`). As benchmarks, the PPA T^3 and the round S^3 Levi–Civita Dirac have spectra exactly symmetric under $\lambda \leftrightarrow -\lambda$, so $\eta_{\text{APS}}^{(3D)} = 0$ (verification: `proofs/aps_zero_t3_s3.py`). On the compact Sol^3 benchmark the local CS integral vanishes, but $\eta_{\text{APS}}^{(3D)}$ branches as 1/0 between Sol-P and Sol-A; Sol^3 is therefore treated as a global spectral branch rather than a local core (Appendix E.13; verification: `proofs/eta_aps_sol3.py`).

B3: torsional-charge rule. The torsional-charge entry appears representatively in S^3 with canonical value

$$N_{\text{top}} = 6r_0^2. \quad (30)$$

This is the reason that pairs containing S^3 are torsional-charge-led in the pairwise cobordism dictionary.

Justify. Starting from the structure constants $C^i_{jk} = (4/r_0) \epsilon^i_{jk}$ of S^3 , one obtains $T^i = \frac{1}{2} C^i_{jk} e^j \wedge e^k$, $\mathcal{C}_3^{\text{tor}} = (12/r_0) \text{vol}_{S^3}$, and hence $\int_{S^3} \mathcal{C}_3^{\text{tor}} = 24\pi^2 r_0^2$. Dividing by the frame-bundle normalisation $1/(4\pi^2)$ ($\pi_3(SO(3)) = \mathbb{Z}$, $T_R = 1$) gives $N_{\text{top}} = 6r_0^2$ (Appendix E.11; verification: `paper04/torsional_charge.py`).

B4: spectral-source rule. Within the spectral family, one must distinguish not only the presence/absence of the observable but also the source mechanism. Nil^3 has a local spectral core supported by a non-zero Levi–Civita spinor CS integral, while Sol^3 may have a global spectral branch sourced by kernel data depending on the compact quotient and spin structure.

Justify. In Nil^3 , $\int_Y \mathcal{C}_3^\sigma = -1/2$ directly fixes $\eta_{\text{APS}}^{(3D)} = +1/2$ (B2). In Sol^3 on the compact mapping-torus benchmark $M_A = T^2 \rtimes_A S^1$ the local CS integral remains $\int_Y \mathcal{C}_3^\sigma = 0$, while constant spinors in the $k = 0$ sector descend only under Sol-P, giving $h = 2/0$ and (with antilinear symmetry $\eta(0) = 0$) $\eta_{\text{APS}}^{(3D)}(\text{Sol-P/Sol-A}) = 1/0$ (Appendix E.13; verification: `proofs/eta_aps_sol3.py`). Hence Nil^3 and Sol^3 share the spectral family but not the same source.

5.3 Pairwise Rules

The pairwise cobordism dictionary furnishes the following rules.

- Pairs containing S^3 tend to be torsional-charge-led. S^3 carries the canonical frame-bundle-normalised torsional-charge core in the second layer (B3) and the bulk side possesses an isotropic activation structure (3-direction CS + triplet degeneracy, R2/R3). However, $S^3 \leftrightarrow \text{Sol}^3$ is torsional-charge-led on Sol-A and a torsional / global-spectral mixed pair on Sol-P.
- Pairs containing Nil^3 tend to be spectrally led. In $T^3 \leftrightarrow \text{Nil}^3$ the local APS core is the subject of comparison; in $\text{Nil}^3 \leftrightarrow \text{Sol}^3$ the “local vs. global” source distinction within the spectral family becomes the subject. This is because Nil^3 carries both an EC slice minimum (R4) and CS-active structure, and is the only Levi–Civita Dirac PPA case with a non-zero local CS core (Appendices E.12, E.13). Sol^3 also carries an EC slice minimum but no local CS core, so its source mechanism is not identical to that of Nil^3 .
- $S^3 \leftrightarrow \text{Nil}^3$ is the representative mixed pair crossing the torsional-charge family ($\mathcal{C}_3^{\text{tor}}$) and the local spectral family ($\mathcal{C}_3^\sigma$). Since the two belong to different 3-form sectors, they cannot be reduced to a single scalar observable (3-form distinction: Appendix E.10).
- $T^3 \leftrightarrow \text{Sol}^3$ is not a uniformly inert minimal pair but a benchmark pair sensitive to compact quotient / spin structure. Sol-A coincides with the trivial spectral profile

of T^3 , while Sol-P raises a global spectral branch with $\eta_{\text{APS}}^{(3D)} = 1$. The scaffold differences (C_{LC}^2 , spin-2 rigidity tag, A_{KK}/K^2) are preserved in both.

5.4 Spectral Source and Geometric Distinctness

A particularly important point about the selection rules is to avoid conflating “belonging to the same family” with “the value being determined by the same source”. Both Nil^3 and Sol^3 may appear in the spectral family, yet Nil^3 is fixed by the local Levi–Civita CS integral while Sol^3 is fixed by global kernel data depending on compact quotient / spin structure. The same spectral family does not imply the same mechanism.

T^3 is flat and inert, while Sol^3 carries a non-vanishing Weyl curvature, biaxial KK splitting, and frame rigidity. Even where Sol-A makes the boundary observable coincide with the T^3 trivial benchmark, the background geometries do not coincide. Conversely, on Sol-P a global spectral branch is added and Sol^3 separates from T^3 already at the second layer. This distinction does not require introducing a third or fourth observable family but reflects differences in the geometric scaffold and global spectral sourcing that support the observables. The selection rules of this paper therefore deliberately separate “compression of observable classes” from “characterisation of background geometry”.

5.5 Summary

By bringing the selection rules to the foreground, the inventory becomes readable as a comparison structure linked to geometry-dependent rules. The next section retains this principal structure and discusses two readings of the Euclidean circle.

6 Secondary Readings of the Euclidean Circle

This section organises two readings of the Euclidean circle S^1 . These provide a reduced description and a thermal description on top of the already constructed Euclidean response dictionary. Crucially, the readings discussed here do not replace the main dictionary; they supply secondary interpretations of the same Euclidean data.

6.1 KK Reduction Reading

In the KK reading we distinguish the geometric response itself from the reduced descriptor that describes it in lower-dimensional language. The effective coefficients and parity-odd channels appearing in the lower-dimensional sector are images of the dictionary already constructed, not the primary objects themselves.

We also fix the geometric setup used in the KK reduction normalisation identity. When extracting coefficients we treat the Euclidean circle as a product/collar direction of length β and apply the CZ identity $\text{NY} = d\mathcal{C}_3^{\text{tor}}$ together with Stokes' theorem on the relative cobordism

$$X^4 = M^3 \times I_\beta, \quad I_\beta = [0, \beta]. \quad (31)$$

The 3D effective coefficient is defined as a functional on a single physical boundary slice $Y = M^3$ that absorbs the volume factor along I_β , so that here $\int_{\partial X^4} \mathcal{C}_3^{\text{tor}}$ refers to this single-copy boundary contribution. The coefficient $k_{3D}^{\text{tor-CS}}$ is therefore the 3D boundary coefficient obtained under inflow/collar normalisation, not a two-endpoint difference along a closed thermal trace. Where the cancellation of the non-zero KK tower is invoked, we assume a real bosonic KK decomposition on an untwisted product circle.

For this reduced-side reading the following three consistency statements are useful:

$$k_{3D}^{\text{tor-CS}} = \frac{1}{2} k_{4D}^{\text{NY}}, \quad Q_{\text{best}} := \frac{\eta(0)}{\Delta k_{\text{matter}}} = 1, \quad k_q \in \mathbb{Z}. \quad (32)$$

The first two give normalisation-free statements about the torsional / matter-coupled parity-odd descriptors, and the last gives the integrality condition for the reduced Chern–Simons level.

The first identity $k_{3D}^{\text{tor-CS}} = (1/2)k_{4D}^{\text{NY}}$ is obtained algebraically as the 3D torsion-CS level after KK reduction along S^1 of the 4D Nieh–Yan action $S_{\text{NY}} = (k_{4D}^{\text{NY}}/8\pi^2) \int d\mathcal{C}_3^{\text{tor}}$ (KK reduction normalisation identity; Appendix E.14; verification: `proofs/kk_normalization.py`). Here, on an untwisted product circle, the Fourier modes n and $-n$ form complex-conjugate pairs and the even weights are determined by n^2 alone, so that the parity-odd residue takes the form $n w(n^2)$. The non-zero tower therefore cancels pairwise and leaves no further parity-odd shift. The second identity $Q_{\text{best}} = 1$ follows from the comparison between the spinor-CS and the APS spectral entry: in Nil^3 one has $\eta(0) = 1$ and $\Delta k_{\text{matter}} = 1$

(matter-minimal inflow audit; Appendix E.15). The third statement $k_q \in \mathbb{Z}$ is the formal quantisation condition required by large gauge transformations on the frame bundle, $\pi_3(SO(3)) = \mathbb{Z}$ (formal quantisation on the frame bundle; Appendix E.16). These are not new observable families of the second layer but consistency statements that arise when the Euclidean response is read in lower-dimensional language.

On the bulk side, the universal Maxwell coefficient

$$K_{\text{Mxw}} = -\frac{L^2}{2r_0^4} \quad (33)$$

is reread as a topology-independent kinetic baseline, while the activated Chern–Simons directions, the uniaxial Higgsing of Nil^3 , and the triplet degeneracy of S^3 are interpreted as topology-dependent channels of the reduced vector sector. The reduced language is therefore a re-reading of the bulk inventory in lower-dimensional form, not something prior to the bulk inventory.

On the boundary side, the torsional inflow action, the APS spectral entry, and the frame-bundle-normalised torsional charge can likewise be mapped to parity-odd descriptors in lower-dimensional language. However, S_{CZ} is written in terms of $\mathcal{C}_3^{\text{tor}}$ while $\eta_{\text{APS}}^{(3D)}$ belongs to Levi–Civita spectral data, so the observable-family distinction is preserved even after projecting onto reduced language. In particular, $\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = 1/2$ and $N_{\text{top}}(S^3) = 6r_0^2$ both admit reduced parity-odd descriptors but are not the same quantity. We do not claim a complete identification between the reduced-side statements and the native geometric quantities.

In this sense the KK language does not replace the response dictionary; it provides a reduced reading of it. The coefficient correspondences and the normalisation chain are collected in Appendix D.

6.2 Matsubara-Style Thermal Reading

In order to overlay a thermal-descriptor reading on the same Euclidean circle, we adopt

$$\beta = L \quad (34)$$

and describe a Matsubara-style thermal reading as the thermal interpretation. Here L is the same length of the Euclidean circle that already appears in the geometric formulation; the thermal reading does not introduce a new circle but assigns thermal meaning to the existing Euclidean data.

Under this interpretation, Maxwell universality, the CS direction-count law, the uniaxial Higgsing of Nil^3 , and the triplet degeneracy of S^3 remain Euclidean structural entries; only the thermal-descriptor language is added to them. Moreover, the Matsubara reading does not break the original product structure $M^3 \times S^1$, so it does not by itself generate a

new parity-odd source. In this sense, what we obtain is a $P = 0$ -protected thermal sector, not a full Lorentzian continuation.

The KK reading and the Matsubara-style thermal reading coexist as two secondary readings of the same Euclidean circle. Coexistence on the same circle, however, does not imply theoretical identity: the former is a lower-dimensional descriptor language and the latter is a thermal descriptor language, and the reduced gauge coefficient, boundary inflow action, phase labels, etc. merely admit both readings.

The thermal reading of this section is deliberately confined to the Euclidean scope. We do not claim a derivation of full finite-temperature field theory, a strict construction of the partition function, a detailed analysis of the Matsubara sum, or real-time transport. The KK–Matsubara coexistence table is collected in [Appendix C](#), and the descriptor separation and normalisation chain in [Appendix D](#).

7 Discussion and Outlook

This section organises the implications of the results and the directions for future work.

7.1 What This Paper Has Fixed

This paper fixes a fairly strong dictionary—rather than the bare minimum—for placing the four Thurston geometries within a single comparison framework. In the first layer the 36-entry bulk inventory closes, making explicit that geometry-dependent activation rules ride on top of Maxwell universality. In the second layer the three observable families spectral / torsional-charge / inflow are clearly separated, and the pairwise cobordism dictionary organises all six pairs into a comparable form.

This organisation positions Nil^3 as the geometry that carries a strict local APS spectral core, S^3 as the geometry that carries a frame-bundle-normalised torsional-charge core, T^3 as the trivial spectral benchmark, and Sol^3 as the geometry that may carry a global spectral branch depending on the compact quotient and spin structure. An important point is that within the spectral family one must still distinguish sources: the value in Nil^3 is fixed by the local Levi–Civita spinor CS, while the spectral entry in Sol^3 is raised by global kernel / spin data. The scaffold layer is needed to describe this difference of source together with the distinctness of the background geometries simultaneously.

By aligning a KK reading and a Matsubara-style thermal reading on the same Euclidean circle we have also made explicit that, with the main dictionary preserved, a reduced description and a thermal description can be added. This shows that the principal result of this paper is the Euclidean response dictionary itself, while the two readings are secondary interpretations of it.

7.2 Future Directions

Direct extensions split into at least the following three lines.

1. **Lorentzian extension.** Lifting the Euclidean response dictionary fixed here to a Lorentzian setting may make it possible to decide which protected / activated structure survives as a real-time observable. In particular, which of parity-odd transport, interface response, and thermal interpretation will withstand analytic continuation is the next-stage question to be made explicit.
2. **Application to phenomenology.** The dictionary is not confined to abstract geometry classification but can serve as a platform that organises which response channels are expected for phenomenological labels such as metastable, rolling, or no-support branches. Using bulk activation and boundary observables together leaves

room for organising geometry dependence that is invisible from a single effective coefficient.

3. **Generalisation of Sol^3 compact-quotient / spin-structure dependence.** This paper makes the boundary spectral entry of Sol^3 strict on a compact mapping-torus benchmark, but how the kernel structure changes for general hyperbolic monodromies and quotient families is left for future work. Making explicit how the local/global dichotomy fixed here generalises across the Sol family is the natural next step for handling borderline spectral entries.

8 Conclusion

In this paper we have constructed a Euclidean geometric response dictionary and the corresponding selection rules for the four Thurston geometries T^3 , Nil^3 , S^3 , and Sol^3 . The principal result is the organisation of the response into a two-layer structure that closes the bulk inventory and the boundary/cobordism inventory across the four geometries.

In the first layer we provided a 36-entry bulk inventory and its selection rules. Here the Maxwell coefficient

$$K_{\text{Mxw}} = -\frac{L^2}{2r_0^4} \quad (35)$$

is common to the four geometries, while the activation conditions for CS, the Higgsing pattern, the EC slice-minimum structure, and the reduced η -mode defect-localisation benchmark branch in a geometry-dependent manner. Nil^3 is organised as a CS+EC activated geometry, S^3 as the most richly CS- and Higgsing-activated geometry, T^3 as an inert benchmark, and Sol^3 as a CS-inert / EC-active geometry with a rigid scaffold.

In the second layer we fixed the three observable families spectral, torsional-charge, and inflow, together with the pairwise cobordism dictionary. In particular, the strict local APS core of Nil^3 , the frame-bundle-normalised torsional-charge core of S^3 , the trivial spectral benchmark of T^3 , and the spin-sensitive global spectral branch on the compact Sol^3 benchmark constitute the centre of the comparison dictionary. This makes explicit the separation of local and global sources within the spectral family. The geometric scaffold layer distinguishes this difference of source from the distinctness of the background geometry.

We have further presented two readings of the Euclidean circle, the KK reduction reading and the Matsubara-style thermal reading. These provide a reduced description and a thermal description respectively, but are not identified theoretically. They coexist as secondary interpretations of the already constructed Euclidean response dictionary.

In conclusion, the present paper offers a Euclidean geometric response dictionary that gives the allowed, protected, and activated responses of the four geometries as a coherent rule set. The principal contribution is the explicit display of how the layers of bulk, boundary, cobordism, and circle readings are organised once the four Thurston geometries are placed within the same comparison framework.

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The AI tools did not determine the scientific claims of this work and were not used to generate or modify research data or evidentiary figures. The author takes full responsibility for the content and for any remaining errors.

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Availability and Contact

Code and Data Availability The code and supplemental data associated with this work are available at:

- Zenodo: [10.5281/zenodo.19775695](https://zenodo.org/record/19775695)
- GitHub: <https://github.com/Muacca/DPPUv7-paper04>

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A Notation and Conventions

This appendix collects the notation conventions used in the body, the shorthand for the four geometries, the terminology of the two-layer structure, and the distinction between three 3-forms that are easily conflated. Since the formulas in the body are written in compact form, the premises of the comparison are made explicit here.

A.1 Geometries and Common Setup

- T^3 : flat three-torus.
- Nil^3 : Heisenberg-type nil geometry.
- S^3 : round three-sphere.
- Sol^3 : solv geometry.

In this paper “the four geometries” always refer to these four. All comparisons are carried out on the Euclidean product structure

$$M^3 \times S^1, \tag{36}$$

with L denoting the length of S^1 . r_0 is the reference length characterising the homogeneous background, and R is the geometric scale appearing in the structure constants of the three-dimensional spatial part.

A.2 Two-Layer Structure and Response Vocabulary

- **first layer**: response that can be defined on the bulk side without introducing a boundary.
- **second layer**: response defined in the boundary / interface / cobordism setting.
- **secondary reading**: an interpretive layer that re-reads, in another language, the already-fixed Euclidean response dictionary.

In the body we further use the following terms.

- **allowed**: the channel is geometrically meaningful.
- **protected**: the channel is not automatically opened even under non-trivial perturbations.
- **activated**: a non-trivial response is raised by the structure of the geometry.
- **minimal**: there is no active family and the geometry sits on the benchmark side.

A.3 Principal Symbols and Representative Values

- K_{Mxw} : Maxwell baseline coefficient.
- S_{CZ} : Callias–Zanelli inflow action.
- $\eta_{\text{APS}}^{(3D)}$: APS spectral invariant.
- N_{top} : frame-bundle-normalised torsional-charge entry.
- C_{LC}^2 : Levi–Civita Weyl scalar.
- spin-2 rigidity tag: scaffold-layer label (`trivial` / `non-rigid` / `rigid`).
- A_{KK} : KK anisotropy marker.
- AX torsion ansatz: $T_{ijk} = (2\eta/\rho) \epsilon_{ijk}$ (DPPU AX-mode convention from paper III (EC) §3; the coefficient $2/\rho$ is fixed by this convention).

Representative benchmark values are listed below. Derivation sketches and verification scripts are referenced in Appendix E.

Quantity	Value	Role	Derivation
K_{Mxw}	$-L^2/(2r_0^4)$	universal bulk baseline	§E.2, §E.6
$\eta_{\text{APS}}^{(3D)}(\text{Nil}^3)$	$+1/2$	strict spectral core	§E.12
$N_{\text{top}}(S^3)$	$6r_0^2$	frame-bundle-norm. torsional-charge core	§E.11
EC slice branch	present for $\text{Nil}^3, \text{Sol}^3$	spin-0 bulk branch	§E.7
$m^2(A_1) = m^2(A_2)$ in Sol^3	$-L^2/(2r_0^4)$	biaxial KK splitting	§E.2, §E.6
$C_{\text{LC}}^2(\text{Nil}^3)$	$4/(3R^4)$	scaffold datum	§E.9
$C_{\text{LC}}^2(\text{Sol}^3)$	$16/(3R^4)$	scaffold datum	§E.9
$k_{3D}^{\text{tor-CS}}/k_{4D}^{\text{NY}}$	$1/2$	NY–CS normalisation identity	§E.14
$Q_{\text{best}} = \eta(0)/\Delta k_{\text{matter}}$	1	normalisation-free invariant	§E.15
k_q (formal)	$\in \mathbb{Z}$	reduced CS level integrality	§E.16

Table 8: Representative benchmark values.

A.4 The Three 3-Forms and Their Connection

We carefully separate the following 3-forms.

- $\mathcal{C}_3^{\text{tor}}$: torsion-CS form.
- $\mathcal{C}_3^\sigma$: spinor CS form.
- $\mathcal{C}_3^{\text{adj}}$: adjoint CS form.

The CZ inflow corresponds to $\mathcal{C}_3^{\text{tor}}$, and the APS spectral entry to $\mathcal{C}_3^\sigma$. In the body these must not be conflated. Conceptually

$$\text{NY} = d\mathcal{C}_3^{\text{tor}}, \quad (37)$$

and $\mathcal{C}_3^{\text{adj}}$ appears at intermediate stages of the APS argument.

A.5 Geometric Scaffold Quantities

The following quantities, although not observable families themselves, are used to record the distinctness of the background geometries.

- C_{LC}^2 : magnitude of the Levi-Civita Weyl curvature.
- spin-2 rigidity tag: scaffold label summarising the zero pattern of the squash / shear Hessian.
- A_{KK}/K^2 : magnitude of the KK anisotropy, with

$$A_{\text{KK}} := \sum_{i=0}^2 (m_i^2 - \bar{m}^2)^2, \quad \bar{m}^2 := \frac{1}{3} \sum_{i=0}^2 m_i^2, \quad K^2 := K_{\text{Mxw}}^2. \quad (38)$$

This scaffold layer does not define an additional observable family, but it is indispensable for distinguishing geometries such as T^3 and Sol^3 that share the absence of a local CS core. In particular, Sol^3 is CS-inert yet carries an EC slice minimum and frame rigidity.

The benchmark values A_{KK}/K^2 can be computed directly from the mass dictionaries:

- T^3 : $m_i^2 = 0$, so $A_{\text{KK}}/K^2 = 0$.
- Nil^3 (uniaxial $m_2^2 = K_{\text{Mxw}}$): $\bar{m}^2 = K/3$, $A_{\text{KK}} = 2(K/3)^2 + (2K/3)^2 = (2/3)K^2$.
- S^3 (triplet $m_i^2 = -2L^2/r_0^4$): all i equal, so $A_{\text{KK}}/K^2 = 0$.
- Sol^3 (biaxial $m_1^2 = m_2^2 = K_{\text{Mxw}}$): $\bar{m}^2 = 2K/3$, $A_{\text{KK}} = (2K/3)^2 + 2(K/3)^2 = (2/3)K^2$.

This symbolic confirmation is performed in `proofs/kk_higgsing.py`.

B Full Response Inventories

This appendix collects the full response inventories that appear in summary form in the body. Where the body compresses the tables for readability, here the tables are gathered together for ease of comparison.

B.1 Bulk Inventory

Entry	T^3	Nil^3	S^3	Sol^3
Maxwell coefficient	$-L^2/(2r_0^4)$	$-L^2/(2r_0^4)$	$-L^2/(2r_0^4)$	$-L^2/(2r_0^4)$
CS direction count	0	1	3	0
KK Higgsing	none	uniaxial	triaxial	biaxial
EC slice minimum	absent	present	absent	present
spin-2 rigidity	non-rigid	non-rigid	non-rigid	rigid
P -linear transport	protected	activated	activated	protected
nonlinear opening	absent	absent	activated	absent
defect-localisation response (η benchmark)	trivial	activated	activated	EC-supported / CS-inert
inhomogeneous bulk response	trivial	activated	activated	CS-inert / EC-active

Table 9: Full bulk inventory.

entry	T^3	Nil^3	S^3	Sol^3
Maxwell	universal	universal	universal	universal
CS count	trivial 0	activated 1	activated 3	trivial 0
KK Higgsing	trivial none	activated uniaxial	activated triaxial	activated biaxial
EC slice	trivial absent	EC-supported present	trivial absent	EC-supported present
spin-2	trivial non-rigid	trivial non-rigid	trivial non-rigid	activated rigid
P linear	protected	activated	activated	protected
nonlinear	trivial absent	trivial absent	activated	trivial absent
eta defect	trivial	activated	activated	EC-supported CS-inert
inhom bulk	trivial	activated	activated	EC-supported CS-inert

Figure 8: 36-entry bulk inventory block diagram. The 9 entries $\times$ 4 geometries of §B.1 are colour-coded as **universal** / **activated** / **protected** / **trivial** / **EC-supported**. In particular, Sol^3 shares CS count 0 with T^3 , but in the EC slice and inhomogeneous response rows it branches to the EC-supported side.

The bulk inventory takes the form “universal baseline + geometry-dependent activation”. Nil^3 and S^3 progress as far as a local parity-odd response on the activated side; T^3 is the minimal / inert benchmark. Sol^3 is CS-inert in the direction count but carries an EC slice minimum and a rigid scaffold, so it is not placed in the same minimal class as T^3 . The row “defect-localisation response (η benchmark)” records not the existence of a bound state in the common reduced 1D η -benchmark but how the localisation links to the additional structure.

B.2 Boundary Inventory

Entry	T^3	Nil^3	S^3	Sol^3
CS interface jump	absent	present	present	absent
APS spectral entry	0	strict +1/2 (local)	0	0/1 (Sol-A / Sol-P)
frame-bundle-norm. torsional charge N_{top}	0	0	$6r_0^2$	0
CZ inflow	absent	activated	activated	absent
edge / interface response	minimal	activated	activated	spin-sensitive
boundary tag	trivial benchmark	APS local core	torsional-charge active	global spectral branch

Table 10: Full boundary inventory.

The strict observable of Nil^3 belongs to the local spectral family, that of S^3 to the torsional-charge family. Furthermore, on the compact mapping-torus benchmark, Sol^3 may carry a global spectral branch. This distinction is the key to understanding the mixed pairs and source differences in the pairwise dictionary.

B.3 Pairwise Cobordism Dictionary

Pair	Class	Dominant observable	Not-comparable note
$S^3 \leftrightarrow T^3$	torsional / trivial	frame-bundle-norm. torsional charge	none
$S^3 \leftrightarrow \text{Nil}^3$	torsional / local spectral	mixed	torsional-charge and local spectral are not the same quantity
$S^3 \leftrightarrow \text{Sol}^3$	torsional / global spectral	spin-sensitive	torsional-led for Sol-A; mixed for Sol-P
$T^3 \leftrightarrow \text{Nil}^3$	trivial / local spectral	spectral	none
$T^3 \leftrightarrow \text{Sol}^3$	trivial / global spectral	spin-sensitive	Sol-A: trivial; Sol-P: global spectral
$\text{Nil}^3 \leftrightarrow \text{Sol}^3$	local / global spectral	spectral	source distinction within the spectral family

Table 11: Full pairwise cobordism dictionary.

In the pairwise table the dominant observable supplies the subject of comparison, and the “not-comparable note” makes explicit either that direct subtraction is meaningless because the observable families differ, or that the source differs even within the same spectral family.

B.4 Strict Normalisation Cores

Together with the boundary observable families, the principal results of this paper are supported by three strict normalisation cores. These appear in the reduced-side consistency statements of §6.1 but do not define new second-layer observable families; they are native quantities that guarantee the internal consistency of the Euclidean response dictionary.

#	Core statement	Descriptor	Status	Verification
C-1	$k_{3D}^{\text{tor-CS}} = \frac{1}{2} k_{4D}^{\text{NY}}$	$\mathcal{C}_3^{\text{tor}}$ inflow normalisation	strict (CZ + Stokes)	§E.14
C-2	$Q_{\text{best}} = 1$	doubly normalisation-free quotient	strict (matter assumption minimal)	§E.15
C-3	$k_q \in \mathbb{Z}$	reduced CS level integrality	formal (single-valuedness)	§E.16

Table 12: Three strict normalisation cores.

B.4.1 KK Reduction Normalisation Identity (C-1)

For the 4D Nieh–Yan action $S_{\text{NY}}^{(4D)} = (k_{4D}^{\text{NY}}/8\pi^2) \int_{X^4} \text{NY}$, applying the CZ identity $\text{NY} = d\mathcal{C}_3^{\text{tor}}$ together with Stokes’ theorem on $X^4 = M^3 \times [0, \beta]$ gives, on the 3D boundary,

$$S_{\text{eff}}^{(3D)} \supset \frac{k_{4D}^{\text{NY}}}{2} \cdot N_{\text{top}}. \quad (39)$$

Hence the coefficient on the normalised basis

$$N_{\text{top}} := \frac{1}{4\pi^2} \int_{M_3} \mathcal{C}_3^{\text{tor}} \quad (40)$$

is fixed as the 3D torsion-CS coefficient,

$$k_{3D}^{\text{tor-CS}} = \frac{1}{2} k_{4D}^{\text{NY}}. \quad (41)$$

Here N_{top} stands for the basis of the frame-bundle-normalised torsional charge; since this paper imposes no extra quantisation condition on r_0 , no general integrality is claimed. The factor $1/2$ originates from $4\pi^2/8\pi^2$ and is independent of r_0 . Furthermore, on the untwisted product circle the parity-odd residue from a real bosonic KK decomposition takes the form $n w(n^2)$, so the contribution of the KK tower ($n \neq 0$) cancels strictly in pairs. Detailed derivation and SymPy verification are in Appendix E.14 and `proofs/kk_normalization.py`.

B.4.2 Matter-Minimal Inflow Audit (C-2)

The quotient $Q_{\text{best}} = \eta(0)/\Delta k_{\text{matter}}$ is formed from the numerator $\eta(0)^{(3D)} = 1$ (§E.12) and the denominator $\Delta k_{\text{matter}} = h_{\text{rep}}/2 = 1$ (Redlich parity shift [9], $h_{\text{rep}} = 2$). Both are *doubly normalisation-free*:

- the numerator is a spectral invariant that does not pass through the 4D APS bridge;
- the denominator depends only on h_{rep} and does not pass through the KK normalisation.

Hence $Q_{\text{best}} = 1$ is a strict invariant independent of normalisation conventions. Details are in Appendix E.15.

Remark 1. The Redlich parity shift is used here only as the standard three-dimensional normalisation unit for the parity-odd level shift. No dynamical fermion field is introduced in the geometric sector, and the comparison is kept at the level of a translation-layer normalisation audit.

B.4.3 Formal Quantisation on the Frame Bundle (C-3)

For the parity-odd effective action $S_{\text{odd}}^{(3D)} = k_q \cdot W[A/\omega]$ on the 3D boundary, the single-valuedness $e^{i\Delta S_{\text{odd}}} = 1$ requires, for the unit winding shift under a large rotation on the frame bundle ($\pi_3(SO(3)) = \mathbb{Z}$),

$$e^{2\pi i k_q} = 1 \iff k_q \in \mathbb{Z}. \quad (42)$$

This gives a formal quantisation condition for the boundary Chern–Simons level associated with the frame-bundle response. In the present work, this condition is used as a boundary-level consistency criterion. Its complete identification with the four-dimensional Nieh–Yan coupling θ_{NY} requires an additional normalization bridge between the four-dimensional Nieh–Yan transgression and the three-dimensional frame-bundle Chern–Simons functional. That bridge is not fixed here. Accordingly, the statement $k_q \in \mathbb{Z}$ should be read as a formal boundary quantisation condition, rather than as a first-principles derivation of the θ_{NY} normalization. Details are in Appendix E.16.

B.4.4 Cross-Audit Summary

C-1, C-2, and C-3 are cross-audited along three independent routes. Here “independent” is restricted to the meaning “no core uses the value of another as input”. Normalisation conventions such as $1/(4\pi^2)$, $1/(8\pi^2)$, and $T_R = 1$ may be shared across cores, but this is a common convention layer rather than circulation. The inputs to each core are listed below.

Core	Input from	Physical quantities	Depends on other core?
C-1: $k_{3D}^{\text{tor-CS}} = k_{4D}^{\text{NY}}/2$	4D NY action + CZ + Stokes	$\mathcal{C}_3^{\text{tor}}$, N_{top} basis	NO
C-2: $Q_{\text{best}} = 1$	spinor APS ($\eta(0) = 1$) + Redlich shift ($h_{\text{rep}} = 2$)	$\mathcal{C}_3^\sigma$	NO
C-3: $k_q \in \mathbb{Z}$	path-integral single-valuedness + $\pi_3(SO(3))$	frame-bundle data	NO

Table 13: No-circulation cross-audit summary.

C-1 belongs to the torsional sector ($\mathcal{C}_3^{\text{tor}}$), C-2 to the spectral sector ($\mathcal{C}_3^\sigma$), and C-3 to formal frame-bundle quantisation; none uses another’s value as input. This is the meaning of “no circulation”. The minimality of the matter assumption appears only in $h_{\text{rep}} = 2$ of C-2, and the complete identification between the native and translation-layer quantities is left outside the scope of this paper.

B.5 Selection-Rule Catalogue

Although the body presents a compact summary, the rule catalogue can be read in the following four clusters. Here R1–R5 are the bulk-rule labels introduced in §5.1, and B1–B4 are the boundary-rule labels of §5.2.

Cluster	Representative rules	Content
bulk universality cluster	R1	geometry-independence of the Maxwell baseline
bulk activation cluster	R2–R5	branching of CS, Higgsing, EC slice minimum, reduced η -benchmark response
boundary family cluster	B1–B4	inflow / local spectral core / torsional-charge / spectral-source distinction
pairwise comparison cluster	§5.3	pairwise dominant observable and mixed pairs

Table 14: Four clusters of selection rules.

This catalogue allows the rule set in the body to be re-read from the viewpoint of “what is universal and what is geometry-dependent”.

C KK vs. Matsubara Coexistence Tables

This appendix tabulates the two readings of the same Euclidean circle. The purpose is to make it visible at a glance that the KK reading and the Matsubara-style thermal reading coexist as secondary readings of the same Euclidean data.

C.1 Basic Coexistence Table

Same Euclidean quantity	KK reading	Matsubara reading	coexist?	Caution
S^1 length L	compactification scale	inverse-temperature circle with $\beta = L$	yes	same circle, two readings
Maxwell baseline	reduced kinetic baseline	thermal stiffness scale	yes	shared coefficient does not imply identity
CS-active structure	reduced active channel count	thermal parity-odd channel count	yes	interpretation differs
S_{CZ}	reduced inflow action	interface free-energy descriptor	yes	not a real-time statement
branch label	reduced branch language	thermal branch language	yes	not a full finite-temperature derivation

Table 15: KK vs. Matsubara coexistence table.

The point of this table is to distinguish “shared coefficient” from “identical theory”. The fact that the same Euclidean quantity admits two descriptor languages does not imply complete agreement of the two theoretical readings.

C.2 Representative Quantities and the Two Readings

Native quantity	KK / reduced reading	Matsubara / thermal reading	Remark
K_{Mxw}	reduced kinetic baseline	thermal stiffness scale	same coefficient, different interpretation
CS direction count	reduced parity-odd channel count	thermal parity-odd channel count	counting survives, meaning differs
$\eta_{\text{APS}}^{(3D)}$	lower-dimensional spectral asymmetry	thermal spectral descriptor	not a transport coefficient by itself
N_{top}	reduced interface CS jump	thermal torsional-sector label	remains frame-bundle-norm. torsional charge
S_{CZ}	reduced inflow functional	thermal interface free-energy descriptor	Euclidean, not real-time

Table 16: Representative quantities and the two readings.

C.3 Interpretive Distinction and Scope

The KK reading provides a lower-dimensional descriptor language and the Matsubara reading provides a thermal descriptor language. The two are based on the same Euclidean circle, but neither claims theoretical identity with the other.

The Matsubara reading preserves the original product structure $M^3 \times S^1$, so within the scope of this paper it furnishes a $P = 0$ -protected thermal sector. The thermal language here therefore does not include the derivation of full finite-temperature field theory, the complete analysis of the Matsubara sum, or real-time transport.

D Computational Notes and Normalisation Chain

This appendix collects the normalisation chain and the descriptor separation used in the body. Only the minimum auxiliary material needed to keep the main dictionary distinct from the secondary readings is recorded.

D.1 Geometric Response and Descriptors

We distinguish the geometric response itself from the descriptor that expresses it in another language. The primary object of the body is always the geometric response dictionary, while the KK reading and the Matsubara-style thermal reading are secondary interpretations attached to it.

- **geometric response**: the primary object of the body.
- **reduced descriptor**: a description in lower-dimensional language.
- **thermal descriptor**: a description in thermal language.

This distinction prevents conflating the response dictionary with the re-reading languages. For example, K_{Mxw} is simultaneously a native geometric quantity, a reduced kinetic baseline, and a thermal stiffness scale; this does not imply that the three are mutually identical. Likewise, $\eta_{\text{APS}}^{(3D)}$ and N_{top} both admit lower-dimensional / thermal descriptors, but as observable families they are distinct.

D.2 Normalisation Chain

The normalisation patch of the CZ boundary term should be read along the chain

$$\text{NY} = d\mathcal{C}_3^{\text{tor}} \implies \int_{X_4} \text{NY} = \int_{M_3} \mathcal{C}_3^{\text{tor}} \implies N_{\text{top}} := \frac{1}{4\pi^2} \int_{M_3} \mathcal{C}_3^{\text{tor}} \implies S_{\text{CZ}}[M_3] = \frac{k_{\text{cl}}}{4\pi^2} \int_{M_3} \mathcal{C}_3^{\text{tor}}. \quad (43)$$

The torsional boundary term, the frame-bundle-normalised torsional charge, and the inflow action are thus seen to be different representations within the same $\mathcal{C}_3^{\text{tor}}$ sector. The fact that the $1/(4\pi^2)$ normalisation corresponds to $T_R = 1$ (frame-bundle Chern-class unit) and that $N_{\text{top}} = 6r_0^2$ in S^3 is derived in Appendix E.11.

The APS spectral entry is not part of this chain. $\eta_{\text{APS}}^{(3D)}$ belongs to the spectral family carried by $\mathcal{C}_3^\sigma$ and must be treated as an observable distinct from N_{top} and S_{CZ} , which are defined directly from $\mathcal{C}_3^{\text{tor}}$. The reason for adopting the three-family taxonomy is precisely that the family distinction is not erased by this normalisation chain.

D.3 Reduced-Side Consistency Statements

Under the KK reduction reading, the lower-dimensional parity-odd language admits the following three consistency statements:

$$k_{3D}^{\text{tor-CS}} = \frac{1}{2} k_{4D}^{\text{NY}}, \quad Q_{\text{best}} := \frac{\eta(0)}{\Delta k_{\text{matter}}} = 1, \quad k_q \in \mathbb{Z}. \quad (44)$$

The first is the correspondence between the 4D Nieh–Yan coefficient and the 3D torsion–Chern–Simons coefficient (KK reduction normalisation identity; Appendix E.14; verification: `proofs/kk_normalization.py`). The second is a normalisation-free quotient using a spectral quantity and a matter shift (matter-minimal inflow audit; Appendix E.15). The third is the integrality condition of the reduced Chern–Simons level (formal quantisation on the frame bundle; Appendix E.16).

In this paper these are not added to the boundary observable family as new entries. Instead, they are positioned as consistency statements that arise when reading the Euclidean geometric response dictionary already given in the body in lower-dimensional language. We do not claim a complete identification between these reduced-side statements and the native geometric quantities.

E Derivations and Proof Sketches

This appendix collects minimal derivation sketches for statements whose results alone are stated in the body and other appendices. The aim is to fix, in a reproducible form, the computations supporting the selection rules and observable benchmarks.

Each subsection follows the structure:

- **Statement:** the assertion in the body.
- **Derivation:** a self-contained derivation sketch.
- **Verification:** reference to a stand-alone Python script that performs the equivalent verification (reproducible).

All verification scripts depend only on the DPPU library bundled in `script/dppu/` together with `sympy`, `numpy`, and `scipy`. Each script is run with `python -X utf8 <path>` and judged from its exit code and a final `RESULT: PASS/FAIL` line.

E.1 Sol³ Structure Constants and Frame Rigidity

Statement (§2.3, §3.3, App. A).

$$C^1_{01} = +\frac{1}{R}, \quad C^2_{02} = -\frac{1}{R}, \quad (45)$$

and these are independent of squash / shear deformations (frame rigidity).

Derivation. From the Sol Lie algebra (DPPU convention $[E_b, E_c] = -C^a_{bc}E_a$) we take the left-invariant coframe $\sigma^0 = dz$, $\sigma^1 = e^z dx$, $\sigma^2 = e^{-z} dy$ and compute the exterior derivatives

$$d\sigma^0 = 0, \quad d\sigma^1 = \sigma^0 \wedge \sigma^1, \quad d\sigma^2 = -\sigma^0 \wedge \sigma^2. \quad (46)$$

For the deformed coframe $e^1 = Rf_1\sigma^1$ with the volume-preserving deformation $f_1 = (1 + \varepsilon)^{2/3}(1 + s)$, $f_2 = (1 + \varepsilon)^{-2/3}/(1 + s)$, one has

$$de^1 = Rf_1 d\sigma^1 = Rf_1 \sigma^0 \wedge \sigma^1 = \frac{e^0 \wedge e^1}{R}, \quad (47)$$

in which f_1 cancels exactly; de^2 is similar. Hence the structure constants are independent of ε, s .

Verification. `proofs/sol3_structure.py`.

E.2 Sol³ Biaxial Higgsing — $m^2(A_i)$ from KK Reduction

Statement (§2.4, §3.5).

$$m^2(A_0) = 0, \quad m^2(A_1) = m^2(A_2) = -\frac{L^2}{2r_0^4} = K_{\text{Mxw}}. \quad (48)$$

Derivation. For the KK photon A_i the modified field strength reads

$$\tilde{F}_{01} = F_{01} + C^a{}_{01}A_a = F_{01} + \frac{A_1}{R}, \quad \tilde{F}_{02} = F_{02} - \frac{A_2}{R}. \quad (49)$$

Substituting into the Maxwell action $-\frac{1}{4}\tilde{F}_{ij}\tilde{F}^{ij}$ and integrating over S^1 yields mass terms for A_1 and A_2 with the same coefficient as K_{Mxw} (same-leg self-coupling). A_0 does not appear in $\tilde{F}$ and remains massless.

Verification. `proofs/kk_higgsing.py`.

E.3 Maxwell Universality — Common Principal Coefficient

Statement (§2.3, R1).

$$K_{\text{Mxw}}^{T^3} = K_{\text{Mxw}}^{\text{Nil}^3} = K_{\text{Mxw}}^{S^3} = K_{\text{Mxw}}^{\text{Sol}^3} = -\frac{L^2}{2r_0^4}. \quad (50)$$

Derivation. The geometry-dependent part enters as the structure-constant correction in the modified field strength

$$\tilde{F}_{ij} = F_{ij} + C^a{}_{ij}A_a, \quad (51)$$

but the quadratic Maxwell principal part always comes from the abelian term $-\frac{1}{4}F_{ij}F^{ij}$. The geometry-dependent structure constants can modify lower-order mass / CS / parity-odd channels, but they do not change the coefficient of $F_{ij}F^{ij}$ itself. Under the common product-circle convention and r_0 scaling, integration over S^1 fixes the principal coefficient to $-L^2/(2r_0^4)$ in all four geometries. Maxwell universality is therefore not a coincidence of the extractor but a consequence of the topology-blindness of the principal symbol.

Verification. `proofs/kk_higgsing.py`.

E.4 Sol³ Chern–Simons Cancellation — Profile-Local Self-Coupling Rule

Statement (§2.3, §3.3, R2).

$$\text{CS}_{\text{Sol}^3} = 0 \quad \text{for profile-local self-couplings} \quad \tilde{F}_{01} = F_{01} + c_{01}(z)A_1, \quad \tilde{F}_{02} = F_{02} + c_{02}(z)A_2. \quad (52)$$

Derivation. The non-abelian cubic correction in Sol³ has two contributions proportional to the structure constants $C^1_{01} = +1/R$ and $C^2_{02} = -1/R$. More generally, even when these are replaced by arbitrary scalar coefficients $c_{01}(z), c_{02}(z)$ under profile-local opening, the CS channel is activated only when the correction has its leg *outside* the index pair $\{i, j\}$ (off-diagonal rule, App. E.5). The corrections in Sol³ have the form “self-referential” with legs inside $\{i, j\}$:

$$A_1 \rightarrow \tilde{F}_{01} : \text{leg } 1 \in \{0, 1\}, \quad A_2 \rightarrow \tilde{F}_{02} : \text{leg } 2 \in \{0, 2\}. \quad (53)$$

Hence the off-diagonal CS coefficient is zero independently of the values or sign relations of c_{01}, c_{02} . The verification script treats c_{01}, c_{02} as independent symbols and confirms that, on substituting the constant scale and $R_{\text{inv}} = 1/R(z)$ from the current Sol³ dictionary as profile-local symbols, the CS coefficient still vanishes. The claim is the profile-local algebraic cancellation of the reduced KK extractor; the full inhomogeneous field equation containing derivative terms such as $R'(z)$ is not derived here.

Verification. `proofs/cs_cancellation.py`.

E.5 Off-Diagonal CS Rule

Statement (§2.3, R2). A CS channel is activated only when the correction lies outside the leg pair $\{i, j\}$ of $\tilde{F}_{ij}$.

Derivation. For KK corrections $\tilde{F}_{ij} = F_{ij} + (\text{coupling})A_k$, the three-point structure expands the CS contribution in terms of ϵ^{ijk} . When $k \in \{i, j\}$ antisymmetry kills ϵ^{ijk} ; only $k \notin \{i, j\}$ contributes a non-trivial off-diagonal CS direction. Classifying the correction types across the four geometries:

Type	Example	Correction	CS
no structure constant	T^3	—	0
off-diagonal	Nil^3	$A_2 \rightarrow \tilde{F}_{01}$ (leg 2 $\notin \{0, 1\}$)	1 direction
off-diagonal (isotropic)	S^3	ϵ_{ijk} -type (all legs off-diagonal)	3 directions
self-referential	Sol^3	$A_1 \rightarrow \tilde{F}_{01}$ (leg 1 $\in \{0, 1\}$)	0

Table 17: Classification of correction types and CS activation.

This shows that what determines CS activation is not “the existence of structure constants” but “the geometric position of the correction type”.

Verification. `proofs/cs_cancellation.py` (off-diagonal count is checked structurally).

E.6 KK Higgsing Four-Type Classification

Statement (§2.4, R3). The KK Higgsing pattern is classified into the four types `none` / `uniaxial` / `triaxial` / `biaxial`, and is realised exactly once each by $T^3, \text{Nil}^3, S^3, \text{Sol}^3$ in this order.

Derivation. Computing the mass matrix $m^2(A_i)$ obtained from KK reduction in parallel across the four geometries gives:

Geometry	Mass dict	Massive count	Type
T^3	$\{\}$	0	none
Nil^3	$\{2 : K_{\text{Mxw}}\}$	1	uniaxial
S^3	$\{0, 1, 2 : -2L^2/r_0^4\}$	3	triaxial
Sol^3	$\{1, 2 : K_{\text{Mxw}}\}$	2	biaxial

Table 18: Massive count for each geometry.

All four types are realised exactly once.

Verification. `proofs/kk_higgsing.py`.

E.7 $\text{Nil}^3/\text{Sol}^3$ EC Slice Minima

Statement (§2.4, §3.2, R4). For the homogeneous EC + NY + Weyl potential of the current DPPU library, an EC-induced slice minimum on the $\eta = V = 0$ section exists in $\text{Nil}^3 \times S^1$ and $\text{Sol}^3 \times S^1$ and is absent in T^3 and round S^3 .

The slice potentials of Nil^3 and Sol^3 are

$$V_{\text{eff}}^{\text{Nil}^3}(R; \alpha) = \frac{4\pi^4 LR}{\kappa^2} - \frac{64\pi^4 L\alpha}{3R}, \quad V_{\text{eff}}^{\text{Sol}^3}(R; \alpha) = 4V_{\text{eff}}^{\text{Nil}^3}(R; \alpha). \quad (54)$$

The factor of 4 is consistent with the geometric ratio $C_{\text{LC}}^2(\text{Sol}^3) = 4C_{\text{LC}}^2(\text{Nil}^3)$ (App. E.9): the symmetric pair of structure constants $C^1_{01} = +1/R$, $C^2_{02} = -1/R$ in Sol^3 produces a Weyl curvature four times that of the single Nil^3 structure constant $C^2_{01} = +1/R$.

For $\alpha < 0$, both share the common stationary radius

$$R_0 = \frac{4\kappa}{\sqrt{3}}\sqrt{-\alpha}, \quad (55)$$

with

$$V_0^{\text{Nil}^3} = \frac{32\sqrt{3}\pi^4 L\sqrt{-\alpha}}{3\kappa}, \quad V_0^{\text{Sol}^3} = 4V_0^{\text{Nil}^3}. \quad (56)$$

Furthermore, the spin-0 block of the full homogeneous Hessian in both geometries shares the determinant

$$\det H_{(\eta,V)} = \frac{262144\pi^8 L^2 \alpha^2}{9} (1 - \kappa^4 \theta_{\text{NY}}^2). \quad (57)$$

At the stationary point $R = R_0$, $\eta = V = 0$ the off-diagonal components satisfy

$$\partial_R \partial_\eta V|_* = 0, \quad \partial_R \partial_V V|_* = 0, \quad (58)$$

so the 3×3 Hessian is block-diagonal as $(R) \oplus (\eta, V)$ (verified symbolically in `paper04/ec_slice_minima.py`). Moreover, the spin-0 block satisfies

$$H_{\eta\eta} + H_{VV} > 0 \quad \text{for} \quad |\kappa^2 \theta_{\text{NY}}| \leq 1. \quad (59)$$

Combined with $\det H_{(\eta,V)} > 0$, the spin-0 block is positive definite, and together with $H_{RR}^{\text{Sol}^3} = 4H_{RR}^{\text{Nil}^3} > 0$ the 3×3 Hessian is positive definite. The EC slice-minimum criterion common to Nil^3 and Sol^3 is

$$|\kappa^2 \theta_{\text{NY}}| < 1 \Rightarrow \text{local minimum}, \quad |\kappa^2 \theta_{\text{NY}}| = 1 \Rightarrow \text{marginal}, \quad |\kappa^2 \theta_{\text{NY}}| > 1 \Rightarrow \text{saddle}. \quad (60)$$

Derivation. `paper04/ec_slice_minima.py` computes the slice potential, the stationary radius, V_0 , and the full Hessian directly from the `Mode.MX`, `NyVariant.FULL` homogeneous potential of the current `dppu` library. T^3 has $V_{\text{eff}}|_{\eta=V=0} = 0$ on the flat zero slice, with no isolated radial branch, while round S^3 has

$$V_{\text{eff}}^{S^3}(r) = -\frac{24\pi^2 Lr}{\kappa^2}, \quad (61)$$

whose non-zero slope precludes a stationary point. The Nil^3 criterion agrees with paper III (EC) [4] §5.2 / Theorem 6; paper 04 extends the same verification to Sol^3 to fix the row of the 4-geometry inventory.

Verification / source. `paper04/ec_slice_minima.py`.

E.8 Reduced 1D η -Mode Defect-Localisation Benchmark

Statement (§3.4). For the reduced 1D AX-torsion ansatz $\eta = \eta(z)$ on a slice-wise homogeneous background, the defect-localisation benchmark is given in Sturm–Liouville form

$$\rho(z) = r(z) \quad (T^3, S^3), \quad \rho(z) = R(z) \quad (\text{Nil}^3, \text{Sol}^3), \quad (62)$$

$$-\frac{d}{dz} \left[K_{\text{geo}}(z) \frac{df}{dz} \right] + M_{\text{geo}}(z)f = Ef, \quad K_{\text{geo}}(z) = M_{\text{geo}}(z) = c_{\text{geo}}\rho(z), \quad (63)$$

with

$$c_{T^3} = c_{\text{Nil}^3} = c_{\text{Sol}^3} = \frac{96\pi^4 L}{\kappa^2}, \quad c_{S^3} = \frac{12\pi^2 L}{\kappa^2}. \quad (64)$$

At least within the Gaussian ρ -dip family, bound states exist analytically for all four geometries.

Derivation. The claim is not the rigorous derivation of the full inhomogeneous field equation but a reduced 1D defect benchmark restricted to the AX torsion mode. The mass side is obtained exactly from the homogeneous engine of the current `dppu` library, while the kinetic side is obtained from the AX contortion gradient under a reduced ansatz with z -direction opening only.

First, as exact homogeneous data,

$$M_{\text{geo}} = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \eta^2} \right|_{\eta=0} = c_{\text{geo}}\rho \quad (65)$$

follows directly from `dppu`. Concretely,

$$M_{T^3} = M_{\text{Nil}^3} = M_{\text{Sol}^3} = \frac{96\pi^4 L}{\kappa^2} \rho, \quad M_{S^3} = \frac{12\pi^2 L}{\kappa^2} \rho. \quad (66)$$

Next, with the AX torsion ansatz

$$T_{ijk} = \frac{2\eta}{\rho} \epsilon_{ijk} \quad (67)$$

(the coefficient $2/\rho$ follows the DPPU AX convention of paper III (EC) §3; App. A.3) and the contortion

$$K_{abc} = \frac{1}{2}(T_{abc} + T_{bca} - T_{cab}), \quad (68)$$

totally antisymmetric AX torsion gives $K_{abc} = \frac{1}{2}T_{abc} \propto (\eta/\rho)\epsilon_{abc}$. In the slice-wise homogeneous ansatz $\eta = \eta(z)$, $\rho = \rho(z)$, the contraction $K_{abc}K^{abc}$ in the EC action gives the mass term $M_{\text{geo}}\eta^2$, while $(\partial_z K_{abc})(\partial_z K^{abc})$ gives the kinetic term $K_{\text{geo}}(\partial_z \eta)^2$.

For the kinetic part, with η alone carrying z -dependence and ρ treated as slice-wise

constant in this reduced ansatz,

$$\partial_z K_{abc} \propto \frac{\partial_z \eta}{\rho} \epsilon_{abc}. \quad (69)$$

Self-contraction yields three factors: the index combinatorics $\epsilon_{abc}\epsilon^{abc} = 6$, the ρ -scaling factor $1/\rho^2$, and the volume integral $\text{Vol}_{\text{geo}}(\rho)$. The gradient part is therefore

$$S_{\text{grad}}^{(2)} = \frac{1}{2} \int dz K_{\text{geo}}(z) (\partial_z \eta)^2, \quad K_{\text{geo}}(z) = \frac{6 \text{Vol}_{\text{geo}}(\rho)}{\kappa^2 \rho^2} \quad (70)$$

(verified in `proofs/eta_kinetic_from_contortion.py` via SymPy). Substituting the volume factor for each geometry yields exactly

$$K_{\text{geo}}(z) = c_{\text{geo}} \rho(z) = M_{\text{geo}}(z). \quad (71)$$

The agreement $K_{\text{geo}} = M_{\text{geo}}$, where the kinetic side (contortion gradient) and the mass side ($\partial_\eta^2 V_{\text{eff}}$ from the homogeneous engine) are computed independently, is a non-trivial fact. The current code base therefore supports not the “exact theorem for a general defect channel” but the “agreement between the exact mass datum and the reduced kinetic datum in the reduced 1D η -benchmark”.

For the Gaussian dip

$$\rho(z) = \rho_0 (1 - A e^{-z^2/w^2}), \quad 0 < A < 1, \quad (72)$$

using the trial function $f_\lambda(z) = e^{-\lambda|z|}$ the Rayleigh quotient becomes

$$\frac{E_{\text{var}}(\lambda)}{M_0} = (1 + \lambda^2) \left[1 - A \lambda w \sqrt{\pi} e^{\lambda^2 w^2} \text{erfc}(\lambda w) \right], \quad M_0 = c_{\text{geo}} \rho_0. \quad (73)$$

For $\lambda \rightarrow 0$ this gives

$$\frac{E_{\text{var}}(\lambda)}{M_0} = 1 - A \sqrt{\pi} w \lambda + O(\lambda^2) < 1, \quad (74)$$

so for sufficiently small $\lambda > 0$ at least one bound state exists. The coefficient c_{geo} cancels out of the ratio E_{var}/M_0 , so the conclusion is common to the four geometries.

A representative numerical scan over $(A, w/\rho_0) = (0.3, 1.0), (0.5, 2.0), (0.7, 1.0)$ for all four geometries gives

$$E_{\text{min}} < M_0 = c_{\text{geo}} \rho_0 \quad (75)$$

in all twelve cases (LOCALIZED).

It is important to note that §E.8 furnishes only “kinematic localisation”. The reduced benchmark above implies the common fact that a radial dip can produce a bound state, but on its own does not determine torsional-charge / parity-odd activation. CS-active structure is required in order to obtain a local parity-odd activated response. Nil^3 carries

the CS activation of R2 together with the EC slice minimum of R4, S^3 carries the CS activation of R2 together with the triplet/nonlinear opening, and Sol^3 carries only the EC slice minimum of R4 without the CS activation of R2, leading to an EC-supported, CS-inert entry; T^3 carries neither. The labels “trivial / activated” in the body table therefore record not the existence of localisation but how this common η -benchmark links to the active structure of the next stage.

Verification. `paper04/eta_defect_coefficients.py`, `paper04/defect_localization.py`.

E.9 Background Weyl Scalar C_{LC}^2

Statement (§4.5, App. A).

$$C_{\text{LC}}^2(T^3) = 0, \quad C_{\text{LC}}^2(\text{Nil}^3) = \frac{4}{3R^4}, \quad C_{\text{LC}}^2(S^3) = 0, \quad C_{\text{LC}}^2(\text{Sol}^3) = \frac{16}{3R^4}. \quad (76)$$

Derivation. From the left-invariant coframe of each geometry one computes the Levi-Civita connection ω^a_b via the Koszul formula and the Riemann curvature $R^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b$, then extracts the Weyl tensor C_{abcd} and evaluates $C_{\text{LC}}^2 = C_{abcd}C^{abcd}$. T^3 is flat, while S^3 is conformally flat, so $C^2 = 0$. Nil^3 is of Bianchi II type and Sol^3 is solv; both have non-zero Weyl values. The fact that Sol^3 is *four times* Nil^3 follows geometrically from the sign-symmetric pair of structure constants $(\pm 1/R)$.

Verification. `proofs/weyl_scalar.py`.

E.10 Distinction of the Three CS 3-Forms

Statement (§4.1, App. A). The 3-forms used in this paper are

- $\mathcal{C}_3^{\text{tor}} = e_a \wedge T^a$ (torsion-CS, EC connection);
- $\mathcal{C}_3^\sigma = \text{Tr}_\sigma(\omega \wedge d\omega + \frac{2}{3}\omega^3)$ (spinor CS, LC, APS final);
- $\mathcal{C}_3^{\text{adj}} = \text{Tr}_{\text{adj}}(\omega \wedge d\omega + \frac{2}{3}\omega^3)$ (adjoint CS, LC, APS intermediate).

The CZ inflow uses $\mathcal{C}_3^{\text{tor}}$, while the APS spectral entry uses $\mathcal{C}_3^\sigma$.

Derivation. $\mathcal{C}_3^{\text{tor}}$ is a primitive of Nieh–Yan: $d\mathcal{C}_3^{\text{tor}} = T^a \wedge T_a + R_{ab} \wedge e^a \wedge e^b = \text{NY}$. By Stokes, $\int_{X^4} \text{NY} = \int_Y \mathcal{C}_3^{\text{tor}}$ supplies the boundary inflow. $\mathcal{C}_3^\sigma$ and $\mathcal{C}_3^{\text{adj}}$ are functionals of the Levi-Civita connection alone, appearing in the geometric part of the APS index theorem (i.e. the $\eta(0)$ computation). However the relation between the two is not simply “factor of two by trace normalisation”. On the Nil^3 benchmark the quadratic trace ratio is 1 but the cubic trace ratio is -2 , and the APS final sign carries the orientation

convention $\eta_{\text{APS}}^{(3D)} = -\int_Y \mathcal{C}_3^\sigma$. The correspondence between $\mathcal{C}_3^\sigma$ and $\mathcal{C}_3^{\text{adj}}$ therefore depends on representation and convention; this paper distinguishes them based on the explicit computation in §E.12.

Hence $\eta_{\text{APS}}^{(3D)}$ (spectral) and N_{top} (frame-bundle-normalised torsional charge), although both “parity-odd boundary quantities”, belong to different 3-form sectors and are not amenable to direct subtraction.

Verification. This subsection is purely classificatory (algebraic trace structure) and requires no script. Structural definitions of the 3-forms are recalled in §4.1 and App. A.4.

E.11 $N_{\text{top}} = 6r_0^2$ for S^3 (Frame-Bundle-Normalised Torsional Charge)

Statement (§4.2, B3).

$$N_{\text{top}}(S^3) = \frac{1}{4\pi^2} \int_{S^3} \mathcal{C}_3^{\text{tor}} = 6r_0^2. \quad (77)$$

Derivation. The derivation splits into two stages. As the geometric part, from the left-invariant structure constants of round S^3 ,

$$C^i{}_{jk} = \frac{4}{r_0} \epsilon^i{}_{jk}, \quad (78)$$

the torsion 2-form is

$$T^i = \frac{1}{2} C^i{}_{jk} e^j \wedge e^k, \quad (79)$$

and

$$\mathcal{C}_3^{\text{tor}} = e_i \wedge T^i = \frac{1}{2} C^i{}_{jk} e_i \wedge e^j \wedge e^k = \frac{12}{r_0} \text{vol}_{S^3}. \quad (80)$$

Using $\text{Vol}(S^3) = 2\pi^2 r_0^3$,

$$\int_{S^3} \mathcal{C}_3^{\text{tor}} = \frac{12}{r_0} \cdot 2\pi^2 r_0^3 = 24\pi^2 r_0^2. \quad (81)$$

The topological-normalisation part takes the Chern-class unit on the frame bundle $\pi_3(SO(3)) = \mathbb{Z}$ in the vector / frame convention $T_R = 1$, fixing the normalisation to $1/(4\pi^2)$ ($8\pi^2$ corresponds to the adjoint $T_R = 2$). Hence

$$N_{\text{top}} = \frac{24\pi^2 r_0^2}{4\pi^2} = 6r_0^2. \quad (82)$$

This paper adopts the vector / frame trace convention $T_R = 1$, which matches the Pontryagin class on the frame bundle to the $1/(8\pi^2)$ normalisation of the 4D Nieh–Yan action (§E.14); under the adjoint convention $T_R = 2$ the 3D normalisation also becomes $1/(8\pi^2)$. We use the single $T_R = 1$ chain throughout.

Since this paper imposes no extra quantisation condition on r_0 , what we obtain is

the value of the frame-bundle-normalised torsional charge $N_{\text{top}} = 6r_0^2$. At the benchmark choice $r_0 = 3$ this gives $N_{\text{top}} = 54$, but integrality for general continuous r_0 requires a separate geometric quantisation condition.

Verification. `paper04/torsional_charge.py`.

E.12 $\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = +1/2$ (Local APS Spectral Core)

Statement (§4.2, B2).

$$\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = +\frac{1}{2}. \quad (83)$$

Derivation. The derivation splits into “confirmation of the low Landau-level structure” and “determination of the value from the CS integral”. Decomposing the Levi–Civita Dirac operator on Nil^3 in the PPA spin structure $(\epsilon_0, \epsilon_1, \epsilon_2) = (+, +, -)$ via the Heisenberg mode decomposition, the condition $p_2 \in \mathbb{Z} + 1/2$ excludes the $p_2 = 0$ sector, and for each p_2 one obtains

$$\mu_n^+ = 2n\omega|k_2| + k_2^2, \quad \mu_n^- = 2(n+1)\omega|k_2| + k_2^2, \quad \omega = |k_2|/r_0. \quad (84)$$

This eigenvalue formula follows directly from the Heisenberg group structure $[E_0, E_1] = (1/r_0)E_2$ of Nil^3 . Fixing each p_2 Fourier mode, (E_0, E_1) generate ladder operators $a, a^\dagger$ following the Heisenberg algebra, and the Levi–Civita Dirac operator can be written as

$$D = \gamma^0 E_0 + \gamma^1 E_1 + ip_2 \gamma^2 + (\text{spin connection}). \quad (85)$$

Diagonalising $D^\dagger D$ in the ladder basis, the Landau-level structure $\mu_n = 2n\omega|k_2| + k_2^2$, $\omega = |k_2|/r_0$ emerges in closed form in each p_2 sector. The pair $\mu_n^\pm$ arises from the different zero-point shifts of spin up/down (verified symbolically by the ladder construction in `proofs/landau_levels_nil3.py`). We use only the closed form within each Heisenberg-decomposed sector; the spectral theorem on the full compact quotient $\Gamma \backslash \text{Nil}^3$ is a separate problem. Hence the $n = 0$ spin-up branch is the unique source of the spectral asymmetry and the kernel under PPA is

$$h = \dim \ker D = 0. \quad (86)$$

The value itself is fixed by constructing the Levi–Civita connection ω^{ab} from the structure constant $C^2_{01} = +1/r_0$ of Nil^3 and combining it with the adjoint generators J_{ab} and the spinor generators $\sigma_{ab} = [\gamma_a, \gamma_b]/2$ to evaluate the Chern–Simons 3-form. Concretely, on the Nil^3 benchmark $r_0 = 3$,

$$\int_Y \text{Tr}_{\text{adj}}(\omega \wedge d\omega) = -\frac{1}{4}, \quad \int_Y \frac{2}{3} \text{Tr}_{\text{adj}}(\omega^3) = +\frac{1}{2}, \quad (87)$$

yielding

$$\int_Y \mathcal{C}_3^{\text{adj}} = +\frac{1}{4}. \quad (88)$$

This intermediate value is reproduced in `paper04/eta_aps_nil3.py`. On the spinor side the quadratic trace ratio is 1 but the cubic trace ratio is

$$\frac{\text{Tr}_\sigma(A^3)}{\text{Tr}_{\text{adj}}(A^3)} = -2, \quad (89)$$

so

$$\int_Y \mathcal{C}_3^\sigma = -\frac{1}{2}. \quad (90)$$

Under the DPPU orientation convention

$$\eta_{\text{APS}}^{(3D)} = -\int_Y \mathcal{C}_3^\sigma. \quad (91)$$

The cubic ratio can be confirmed directly with explicit generators. In the 3D Euclidean gamma algebra,

$$\sigma_{ab} = \frac{1}{2}[\gamma_a, \gamma_b] = i\epsilon_{ab}^{c} \gamma_c, \quad (92)$$

so for the spinor basis $T_i = i\gamma_i$,

$$\text{Tr}_\sigma(T_i T_j T_k) = 2 \epsilon_{ijk}, \quad (93)$$

while for the adjoint basis J_i ,

$$\text{Tr}_{\text{adj}}(J_i J_j J_k) = -\epsilon_{ijk}. \quad (94)$$

Substituting the same connection coefficients $A = a^i T_i \leftrightarrow a^i J_i$,

$$\frac{\text{Tr}_\sigma(A^3)}{\text{Tr}_{\text{adj}}(A^3)} = -2. \quad (95)$$

Therefore

$$\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = +\frac{1}{2}, \quad (96)$$

and the APS formula gives

$$\eta(0) = 2\eta_{\text{APS}}^{(3D)} - h = 1. \quad (97)$$

The Landau-level analysis fixes “which branch carries the asymmetry” and “ $h = 0$ in PPA”; the value is fixed by the Levi–Civita CS integral and the spinor / adjoint cubic trace ratio.

The EC correction $\delta = -\kappa_{\text{tor}}/(4r_0)$ is a uniform shift only and produces no zero crossing in the standard parameter range, so $\eta^{\text{EC}}(0) = \eta^{\text{LC}}(0)$.

As benchmarks, the PPA spin structure on T^3 and the Levi–Civita Dirac on round

S^3 both have eigenvalues exactly paired under $\lambda \leftrightarrow -\lambda$ with $h = 0$, so $\eta_{\text{APS}}^{(3D)} = 0$. In Sol^3 , by contrast, the local CS integral itself vanishes but a global spectral branch dependent on the compact quotient and spin structure may arise. This is made explicit in the next subsection as a benchmark.

Verification. The strict Nil^3 value: `paper04/eta_aps_nil3.py`. The T^3/S^3 benchmark zero: `proofs/aps_zero_t3_s3.py`.

E.13 $\eta_{\text{APS}}^{(3D)}(\text{Sol}^3)$ on the Compact Mapping-Torus Benchmark

Statement (§4.2, B4). On the compact Sol^3 benchmark

$$M_A = T^2 \rtimes_A S^1, \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad (98)$$

together with the two benchmark spin structures Sol-P / Sol-A tracked within DPPU,

$$\eta_{\text{APS}}^{(3D)}(\text{Sol-P}) = 1, \quad \eta_{\text{APS}}^{(3D)}(\text{Sol-A}) = 0. \quad (99)$$

“Sol-P” is base periodic / fibre periodic, “Sol-A” is base anti-periodic / fibre periodic. This is not a universal Sol^3 value but a global spectral branch that depends on this compact quotient and spin-structure choice.

Derivation. From the structure constants

$$C^1_{01} = +\frac{1}{r_0}, \quad C^2_{02} = -\frac{1}{r_0} \quad (100)$$

of Sol^3 one constructs the Levi–Civita connection. Both adjoint and spinor Chern–Simons 3-forms then satisfy

$$\int_Y \mathcal{C}_3^{\text{adj}} = 0, \quad \int_Y \mathcal{C}_3^\sigma = 0, \quad (101)$$

so the local CS integral alone does not fix the APS value in Sol^3 .

In a left-invariant orthonormal frame the zero-order spin connection of the Dirac operator cancels exactly, leaving

$$D_{\text{Sol}} = \gamma^a E_a. \quad (102)$$

Decomposing on the mapping torus by fibre Fourier mode $k \in \Lambda^*$, the zero-mode equation in the $k \neq 0$ sector reduces locally, by a $U(1)$ rotation, to

$$(\partial_t \pm m_k(t))\phi = 0, \quad m_k(t) > 0, \quad (103)$$

so for either periodic or anti-periodic boundary condition along the base no zero modes

arise from $k \neq 0$.

In the remaining $k = 0$ sector only constant spinors are candidates. In Sol-P (base periodic) the two constant spinors descend to the quotient, giving

$$h(\text{Sol-P}) = 2, \quad (104)$$

while in Sol-A (base anti-periodic) they do not descend, giving

$$h(\text{Sol-A}) = 0. \quad (105)$$

The value $h = 2$ in Sol-P refers to the standard $\text{Spin}(2)$ lift of the monodromy $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \in SL(2, \mathbb{Z})$ (the benchmark choice inherited by DPPU from paper III (EC)). A admits two spin lifts; another lift can give $h(\text{Sol-P}) = 0$. The Sol-P / Sol-A values here are compact mapping-torus values fixed under this benchmark spin lift, not universal spectral entries of Sol³.

Setting

$$T = i\sigma_2 K \quad (106)$$

gives $T\sigma_j T^{-1} = -\sigma_j$, and since E_a are real vector fields $TDT^{-1} = -D$. T is antiunitary, so for $D\psi = \lambda\psi$ ($\lambda \in \mathbb{R}$) one has $D(T\psi) = -\lambda(T\psi)$: the non-zero spectrum is paired by $\lambda \leftrightarrow -\lambda$. Hence the antilinear time-reversal symmetry yields $\eta(0) = 0$ for both spin structures. By the APS definition

$$\eta_{\text{APS}}^{(3D)} = \frac{\eta(0) + h}{2} \quad (107)$$

we therefore obtain

$$\eta_{\text{APS}}^{(3D)}(\text{Sol-P}) = 1, \quad \eta_{\text{APS}}^{(3D)}(\text{Sol-A}) = 0. \quad (108)$$

The spectral entry of Sol³ in the present benchmark is therefore not a local Levi–Civita CS core (as in Nil³) but a global kernel branch supported by the compact quotient / spin structure. More general hyperbolic monodromy quotients can change this global branch further; in this paper we adopt the M_A benchmark as the reference.

Verification. `proofs/eta_aps_sol3.py`.

E.14 KK Reduction Normalisation Identity

Statement (§6.1, App. D.3).

$$k_{3D}^{\text{tor-CS}} = \frac{1}{2} k_{4D}^{\text{NY}}. \quad (109)$$

Derivation. Write the DPPU 4D NY action with an independent symbol k_{4D}^{NY} :

$$S_{\text{NY}}^{(4D)} = \frac{k_{4D}^{\text{NY}}}{8\pi^2} \int_{X^4} \text{NY}. \quad (110)$$

Applying the CZ identity $\text{NY} = d\mathcal{C}_3^{\text{tor}}$ together with Stokes' theorem on $X^4 = M^3 \times [0, \beta]$ gives

$$\int_{X^4} \text{NY} = \int_Y \mathcal{C}_3^{\text{tor}} = 4\pi^2 \cdot N_{\text{top}} \quad (111)$$

(the last equality is the normalisation of §E.11). Substituting,

$$S_{\text{eff}}^{(3D)} \supset \frac{k_{4D}^{\text{NY}}}{8\pi^2} \cdot 4\pi^2 \cdot N_{\text{top}} = \frac{k_{4D}^{\text{NY}}}{2} \cdot N_{\text{top}}. \quad (112)$$

The coefficient on the normalised basis

$$N_{\text{top}} := \frac{1}{4\pi^2} \int_Y \mathcal{C}_3^{\text{tor}} \quad (113)$$

is identified with the 3D torsion-CS coefficient, giving

$$k_{3D}^{\text{tor-CS}} = \frac{k_{4D}^{\text{NY}}}{2}. \quad (114)$$

The factor $1/2$ comes from $4\pi^2/8\pi^2$ and is independent of r_0 . Here N_{top} is used as the basis of the frame-bundle-normalised torsional charge; this paper does not assume integrality for general continuous r_0 . Furthermore, on an untwisted product circle the real bosonic KK decomposition has Fourier modes n and $-n$ as complex-conjugate pairs; masses and even weights are determined by n^2 alone. The reflection $\tau \mapsto -\tau$ on the circle simply swaps these two modes while preserving the 4D kinetic operator, so the parity-odd residue takes the form $n w(n^2)$ in general, and the contribution of the KK tower ($n \neq 0$) cancels strictly in pairs.

Verification. `proofs/kk_normalization.py`.

E.15 Normalisation-Free Quotient $Q_{\text{best}} = \eta(0)/\Delta k_{\text{matter}} = 1$

Statement (§6.1, App. D.3).

$$Q_{\text{best}} := \frac{\eta(0)}{\Delta k_{\text{matter}}} = 1. \quad (115)$$

Derivation. The numerator $\eta(0)^{(3D)} = 1$ follows from §E.12. The denominator $\Delta k_{\text{matter}} = h_{\text{rep}}/2 = 1$ comes from the Redlich parity shift [9] with representation dimension $h_{\text{rep}} = 2$ (DPPU minimal matter assumption); it is theory-derived. Both are *doubly normalisation-free*:

- the numerator is a spectral invariant that does not pass through the 4D APS bridge;
- the denominator depends only on h_{rep} and does not pass through the KK normalisation.

Hence the quotient Q_{best} is a strict invariant independent of normalisation conventions.

Note. $h_{\text{rep}} = 2$ corresponds to the minimal two-component matter convention used since paper III (EC) and supplies the standard unit of the Redlich parity shift in lower-dimensional language. The identification of $Q_{\text{best}} = 1$ with $k_q = 1$ requires a separate identification step (the spectral $\leftrightarrow$ CS-level dictionary in the DPPU APS convention) which lies outside the scope of this appendix.

Verification. This subsection is the algebraic identity obtained by substituting $\eta(0) = 1$ from §E.12 and $\Delta k_{\text{matter}} = 1$ from $h_{\text{rep}} = 2$.

E.16 CS-Level Integrality $k_q \in \mathbb{Z}$ (Single-Valuedness)

Statement (§6.1, App. D.3).

$$k_q \in \mathbb{Z} \quad [\text{from single-valuedness of } e^{iS_{\text{odd}}}] . \quad (116)$$

Derivation. Write the parity-odd effective action on the 3D boundary $Y = M^3$ as $S_{\text{odd}}^{(3D)} = k_q \cdot W[A/\omega]$. Here W is a parity-odd functional, normalised so that the unit winding under a large frame rotation (frame bundle $\pi_3(SO(3)) = \mathbb{Z}$) gives $\Delta W = 1$. In the 2π -normalised formulation,

$$S_{\text{odd}}^{(3D)} = k_q \cdot (2\pi \cdot W), \quad \Delta W|_{n=1} = 1, \quad (117)$$

so the unit winding shift is $\Delta S_{\text{odd}}|_{n=1} = 2\pi k_q$. Single-valuedness of the path-integral measure $e^{i\Delta S_{\text{odd}}} = 1$ then gives

$$e^{2\pi i k_q} = 1 \iff k_q \in \mathbb{Z}. \quad (118)$$

Note. The homotopy input $\pi_3(SO(3)) = \mathbb{Z}$ used here is the same frame-bundle datum as in the normalisation of §E.11; it corresponds to the large-rotation version of standard 3D Chern–Simons quantisation. The correspondence with the DPPU native θ_{NY} normalisation is not fixed here, and $k_q \in \mathbb{Z}$ itself is a formal statement.

Verification. This subsection is a formal argument based on the single-valuedness of the path-integral measure and requires no additional numerical verification.

E.17 Summary Table

Statement	Body location	Verification	Status
Sol ³ structure constants + frame rigidity	§2.3, App. A	proofs/sol3_structur e.py	exact (SymPy)
Sol ³ rigidity: $\partial_{\varepsilon,s}^2 V = 0$ at $\varepsilon = s = 0$	§4.5, §E.1	proofs/sol3_structur e.py	direct Hessian check
Sol ³ biaxial Higgsing $m^2(A_i)$	§2.4, §3.5	proofs/kk_higgsing.p y	exact (KK extractor)
Sol ³ CS = 0 cancellation	§2.3, §3.3	proofs/cs_cancellati on.py	exact within profile-local KK ansatz
Maxwell universality K_{Mxw}	§2.3, R1	proofs/kk_higgsing.p y	common principal coefficient
Off-diagonal CS rule (R2)	§5.1	proofs/cs_cancellati on.py	classification
4-type Higgsing one-to-one	R3	proofs/kk_higgsing.p y	exhaustive
EC slice minima: Nil ³ /Sol ³ present; T^3/S^3 absent	§2.4, §3.2, R4	paper04/ec_slice_min ima.py	exact homogeneous Hessian criterion
K_{geo} from AX contortion gradient	§E.8	proofs/eta_kinetic_f rom_contortion.py	independent mass/kinetic derivation
reduced 1D η -benchmark: $K_{\text{geo}} = M_{\text{geo}} = c_{\text{geo}}\rho(z)$	§3.4	paper04/eta_defect_c oefficients.py; paper04/defect_local ization.py	exact (4 geom.); variational + numerical (12/12 LOCALIZED)
C_{LC}^2 (4 geometries)	§4.5, App. A	proofs/weyl_scalar.p y	exact (SymPy)
3-form distinction	§4.1, App. A	algebraic; no script	classification
$N_{\text{top}}(S^3) = 6r_0^2$	§4.2	paper04/torsional_ch arge.py	geometry-derived + frame-bundle normalisation
$\eta_{\text{APS}}^{(3D)}(T^3) = \eta_{\text{APS}}^{(3D)}(S^3) = 0$	§4.2, §5.2	proofs/aps_zero_t3_s 3.py	exact benchmark; symmetric LC spectra

(continued on next page)

Statement	Body location	Verification	Status
$\eta_{\text{APS}}^{(3D)}(\text{Nil}^3) = +1/2$	§4.2	<code>paper04/eta_aps_nil3.py</code>	local LC CS + spinor trace derivation
Heisenberg Landau levels for Nil^3 Dirac	§E.12	<code>proofs/landau_levels_nil3.py</code>	symbolic ladder construction
$\eta_{\text{APS}}^{(3D)}(\text{Sol-P} / \text{Sol-A}) = 1/0$	§4.2 , §5.2	<code>proofs/eta_aps_sol3.py</code>	global / kernel-sourced compact benchmark
$k_{3D}^{\text{tor-CS}} = k_{4D}^{\text{NY}}/2$	§6.1	<code>proofs/kk_normalization.py</code>	strict (CZ + Stokes)
$Q_{\text{best}} = 1$	§6.1	algebraic; §E.12 + Redlich	strict; normalisation-free
$k_q \in \mathbb{Z}$	§6.1	formal; single-valuedness	formal

Table 19: Summary of derivations and verifications.

All verification scripts are placed in `scripts/proofs/` or `scripts/paper04/`, and are self-contained, depending only on the `dppu/` library bundled with the paper.

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